

GE Healthcare

LightSpeed 4.X Installation Manual

Book 2 of 2: Electrical Installation

Operating Documentation



2341957-100
Revision: 07

Effectivity

The information in this service manual applies to the following LightSpeed 4.X CT systems:

—Lightspeed 16 (16-slice MDAS)

Retain this installation manual permanently with your CT System. Any other revisions of this installation manual may not exactly match your system.

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Chapter 1

Electrical Introduction



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 5](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Introduction

Use the continuity and ground checks to verify the system power connections have not shorted to ground and that the ground and neutral connections are intact.

Section 2.0

Review Mechanical Hand Off Material

Complete the Mechanical Hand Off checklist:

- ☐ All options were installed. If not, contact your install specialist.
- ☐ Check for short ships.
- ☐ Review cable connections with mechanical team.
- ☐ Complete mechanical vendor evaluation.

Section 3.0

Electrical Installation Block Diagram

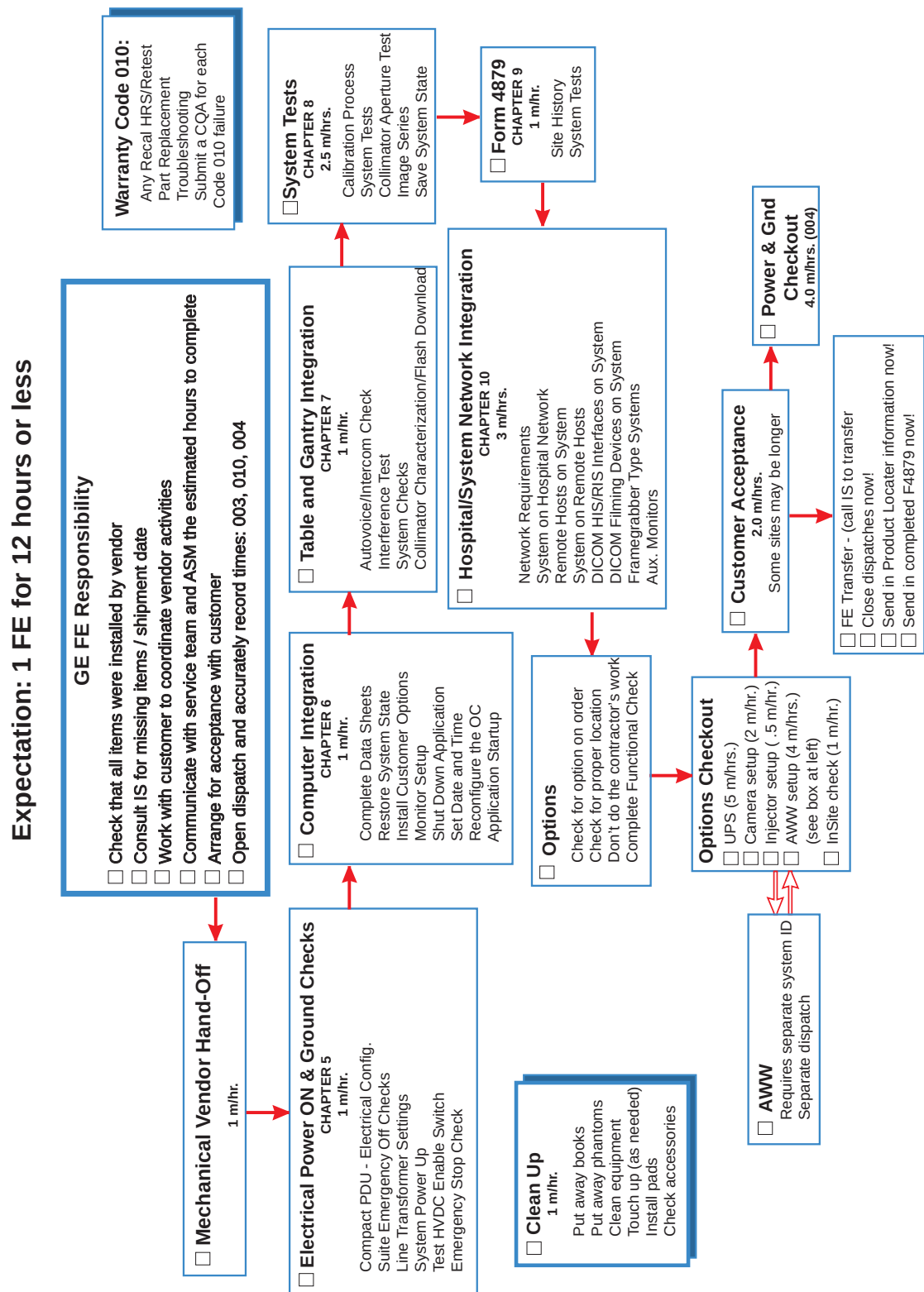


Figure 1-1 Electrical Installation Block Diagram

Section 4.0

Required FE Common Tools and Supplies

4.1 FE Calibration and Service Tool List

Note: Items with a “check” (4) are included in the Install Support Kit.

Special CT Tools used for mechanical alignments:

- | | |
|---|---|
| 4 5mm hex bit for 3/8" drive 6" long
(Snap-On FAML5E or equivalent) | 4 10mm open-end thin wrench
(Snap-On SRSM10 or equivalent) |
| 4 10mm hex bit for ratchet wrench (ball end)
(Snap-On FABM10E or equivalent) | 4 21mm open-end thin wrench
(Snap-On LTAM2124 or equivalent) |
| 4 2.5mm Allen hex bit for 1/4" | 4 14mm ball hex socket bit for 3/8" drive |
| • Standard FE Tool Kit | 4 14 mm hex socket bit for 1/2" drive |

4.2 Electrical Tools

These tools must be calibrated yearly:

- Fluke 87 DVM or equivalent
- Clamp on amp meter

4.3 Image Quality Calibration Tools

- GE Performance Phantom
- QA Phantom (2206352)
 - 20cc (2144721)
 - 40cc (2144721-2)
 - Phantom Holder

4.4 Detector Service Tools

A DAS/Detector Service Kit (PN 2344539) is included in the Install Support Kit.

4.5 Optional Tools

Needed for warranty recalibration:

- GE HV Bleeder
- Scope with 10X probes
- Beckman CT231 clamp-on Amp probe: 46-194427P228 (Fits up to #2 size wire)
- Beckman CT232 clamp-on Amp probe: 46-194427P270 (Fits #1 size wire and larger)
- Scope Probe-to-bleeder Cable 46-219921G1 (Belden #8422 cable, 30 ft)
- Polaroid type 52 film and developer
- Radial dial indicator (mm or inches) and mounting bracket
- Caliper Dial indicators (mm or inches)
- Ground Rod Tester Clamp on Amp Probe
- Torque Wrench kit

4.6 Safety Materials

Note: Items with “checks” (4) are included in the Install Support Kit

- 4 Lockout/Tagout kit, or equivalent
- 4 Nitrile Rubber Gloves
- Safety Glasses

4.7 Cleanliness

Any dirt on the surface increases leakage current on the filter or converter cards and causes the DAS to fail the drift spec. Wear Nitrile gloves (part number 2207303-6 [large] and 2207303-7 [extra large]) when you handle the DAS, because fingerprints on the board can cause problems during humid conditions. Use only clean, new Nitrile gloves. Do not use hospital grade gloves.



NOTICE NEVER USE AN ERASER TO CLEAN ANY PART OF THE DAS.
DUST COVERS REQUIRED for installations on construction sites.

Section 5.0

Requirements/Assumptions

- This procedure will be performed by a LightSpeed 4.X trained engineer.
- All stations in a suite must have the same SUITE ID.
- You need the name of all hosts in the suite and their corresponding Internet/Ethernet numbers.
- You need the Internet (IP) addresses the first time you execute a reconfig on the system.
 - When you connect the system to a network, contact the system administrator to obtain the IP addresses for all the computers in the suite.
 - You also need an IP address for each gateway (second) ethernet board in any OC or IC.
 - You can use the default internet number on stand-alone systems (stand-alone = not connected to any network).
- For more detailed information and instructions regarding Network Integration, see [Chapter 4 on page 69](#).

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Chapter 2

Electrical Integration and Safety Verifications



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 5](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Electrical Power ON & Ground Checks

1.1 Compact PDU - Electrical Configuration



WARNING



THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

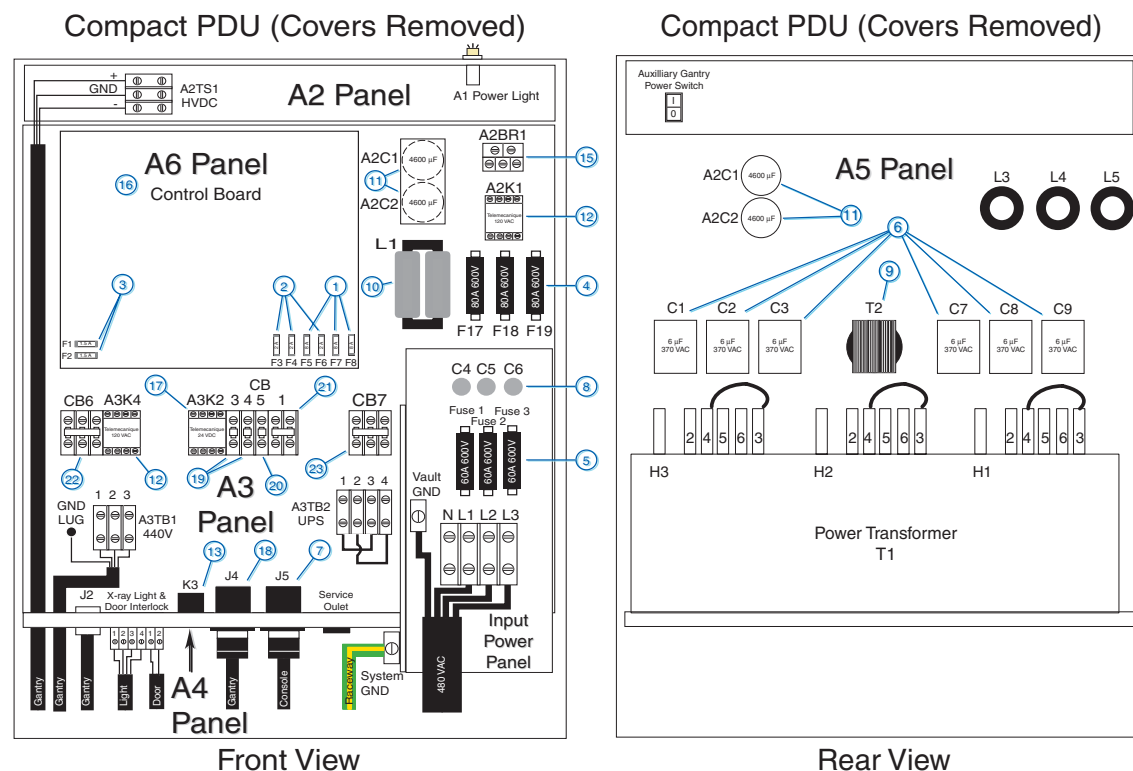


Figure 2-1 Compact PDU

1.1.1 Required Tools

- Multimeter with a rating of at least 1000 volts
- Multimeter leads with a rating of at least 1000 volts

1.1.2 Initial PDU Configuration



WARNING

THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

1.1.2.1 Circuit Breakers

Set all circuit breakers to OFF [CB 1-7]

1.1.2.2 Relay Board

- 1.) Set SW1 and SW2 to the normal position.
- 2.) Set the Room Light jumper (JP1) to meet hospital requirements (refer to [Figure 2-2](#)).
 - Position A enables the room warning light during X-Ray only.
 - Position B enables the room warning light as soon as HVDC turns on.

Note:

Use position B unless the hospital requires otherwise.

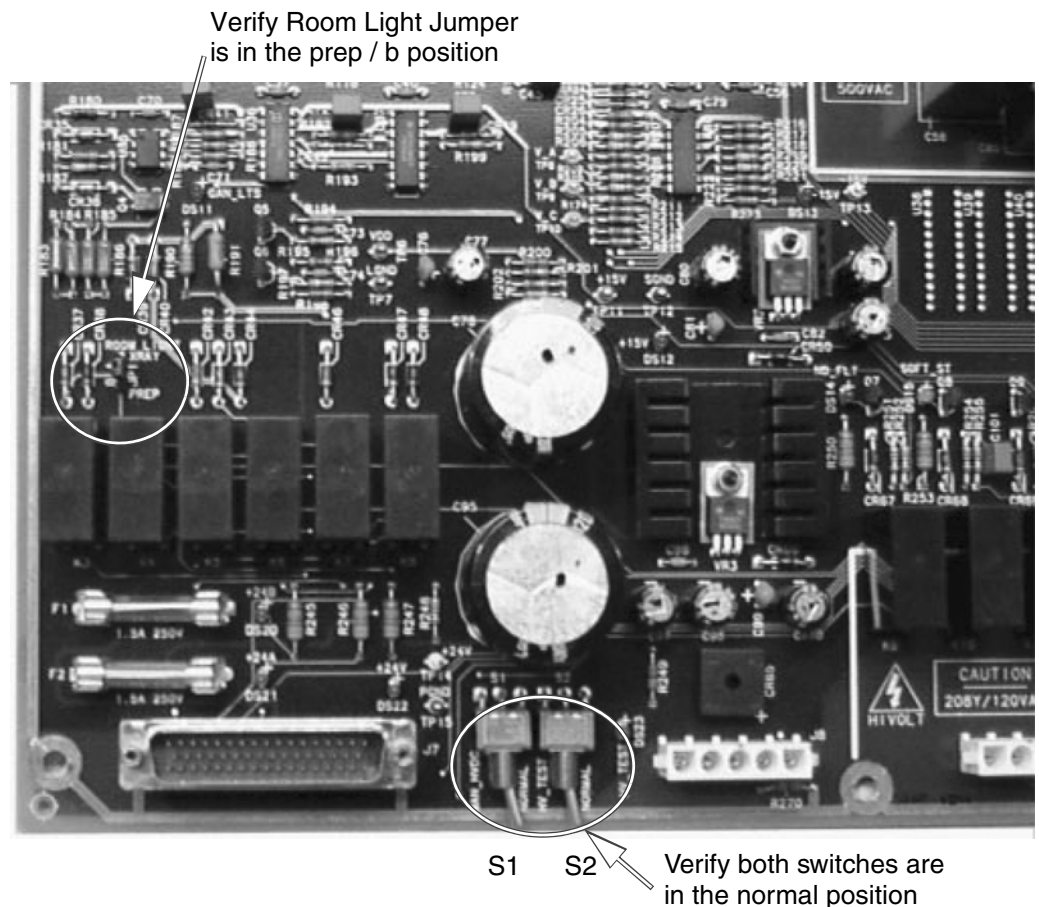


Figure 2-2 CPDU Control Board (A6 Panel)

1.1.2.3 Power Switches

Turn OFF all power switches on all subsystems

- STC
- DAS
- Table
- Console

1.1.2.4 Hardware and Connection Check

Use this step to check mechanical connections and tighten anything that may have shaken loose during shipment. Verify **all** hardware and connections in the PDU are securely fastened.

- Fuses
- J4, J5
- A6 Panel (control board)

1.1.2.5 Covers

Install, or verify the presence of, all the lexan safety covers.

1.2 Suite Emergency Off Checks



WARNING

VERIFY ALL PERSONNEL HAVE CLEARED THE SYSTEM BEFORE TURNING ON WALL POWER.

- 1.) Turn wall power ON to the PDU.
- 2.) **Press the suite emergency off button and verify it turns off wall power to the PDU.**
[Typically, this red palm button is located on the wall close to the console, within the scan suite.]
- 3.) Verify that all “Emergency Off” button are working properly.
- 4.) Leave power “OFF”

1.3 Line Transformer Settings

1.3.1 Requirements

- 1.) The PDU is shipped configured for 480VAC.
- 2.) Complete only if your site uses a voltage other than 480VAC.
- 3.) If PDU is configured for 480VAC, go to 1.4. Otherwise, proceed to [Section 1.3.2](#).



WARNING



MAKE SURE THE MAIN WALL POWER HAS BEEN TURNED OFF, TAGGED AND LOCKED BEFORE CHANGING TAPS. FAILURE TO DISCONNECT POWER AT MAIN INPUT MAY RESULT IN ELECTROCUTION. TURN OFF WALL POWER TO CONNECT OR MOVE METER LEADS, OR TO REMOVE OR INSTALL COVERS.

1.3.2 Line Input Conditions

- 1.) Remove metal cover over right A3 panel.
- 2.) Monitor the No Load Line to Line Voltage at A3 L1, L2, L3, during the workday. Do not record this data during “brown out” conditions.
- 3.) Remove the PDU’s back cover (see [Figure 2-1](#) for a back view of the PDU with cover off)

- 4.) After you determine the nearest nominal line, verify the tap connections match (refer to [Table 2-1](#) and [Figure 2-3](#) for tap locations).

Transformer Taps and Jumpers

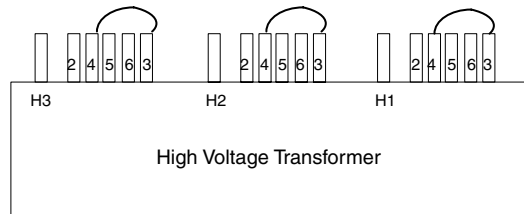


Figure 2-3 PDU Tap Positions (Rear)

Note: **Taps should be shipped as shown for 480 VAC only. For all others, you must move the taps. The tap check should be completed by the mechanical installer.**

- 5.) Verify that the No Load Line to Line Voltage never falls outside the corresponding minimum and maximum values listed in [Table 2-1](#).
- 6.) Use a 0-750 AC voltmeter of $\frac{3}{4}\%$ accuracy to measure the line-to-line voltages at A3 L1, L2, and L3.
- Verify the highest line-to-line voltage does not exceed 1.02 times the lowest voltage.
 - **Example:** If the lowest voltage equals 474, the highest voltage should not exceed $474 \times 1.02 = 483.5$ volts.



THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

- 7.) Replace the metal cover over the right A3 panel.

NO LOAD Line to Line Voltages		TAP CONNECTIONS (All 3 phases must have same the configuration)		
Nominal	Maximum Range (8%)	Phase A Connection	Phase B Connection	Phase C Connection
480V*	442 to 518*	3-4*	3-4*	3-4*
460V	423 to 497	3-5	3-5	3-5
440V	405 to 475	3-6	3-6	3-6
420V	386 to 454	2-4	2-4	2-4
400V	368 to 432	2-5	2-5	2-5
380V	350 to 410	2-6	2-6	2-6
* Factory Default				

Table 2-1 PDU 2269902 Line Tap Connections

1.4 System Power Up

- 1.) Turn OFF wall power.
- 2.) Turn ON all PDU A3 circuit breakers. [CB1-7]



Verify all personnel have cleared the system before you turn on wall power.

- 3.) Turn ON wall power. (A1 breaker)

SUB-SYSTEM POWER-UP

- 1.) Turn ON switch S3 in the table (120vac 24hr power).
- 2.) Turn ON the gantry 120VAC enable switch on the STC. Tube fans turn on.
- 3.) Leave AXIAL DRIVE ENABLE OFF.
- 4.) Leave HV DC ENABLE OFF.
- 5.) Turn ON the DAS power switch located on the outside of the DAS power supply assembly.
- 6.) Turn ON Table switches S1 and S2.
- 7.) Turn ON console power.
- 8.) Make certain that the switch on the back of the PDU is ON.
- 9.) Turn ON HEMRC Power ON.
- 10.) Verify that the gantry 120VAC is ON at the Gantry Power Pan.

1.5 Test HVDC Enable Switch

Note: Console must be booted and operational.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select TABLE/GANTRY INTEGRATION.
- 5.) Select HV DC ENABLE SWITCH.
- 6.) Select RUN.
- 7.) Select the following parameters in the Rotor Functional window:



PARAMETER	MX200
Test Selection:	HEMRC
Rotor Speed:	LOW
Rotor Loop:	OPEN
Rail Diff Voltage:	Normal
Test Duration:	10 seconds
OBC Service Input:	Disabled

Table 2-2 Rotor Functional Window Parameters

- 8.) Verify all personnel have cleared the system, *before* you enable the HVDC.
- 9.) Verify the following:
 - a.) HVDC enable switch on the STC is ON
 - b.) "HVDC ENABLE ON" light on the STC is ON
 - * DS4 should NOT be blinking
 - * Press S1 to reset
- 10.) Select RUN. If the system asks for a diagnostic download at this time, answer YES, and wait 90 seconds.
- 11.) Verify the "HVDC BACKUP CONTACTOR CLOSED" light on the STC status display turns ON.

- 12.) When the test finishes, verify the system displays the following message along the bottom of the screen: Diagnostics have completed with zero errors reported.



Figure 2-4 STC Service Switches

VERIFY/PERFORM

- 1.) Turn OFF the HVDC enable switch on the gantry STC.
- 2.) Select RUN to run the test again.
- 3.) Verify the test aborts.
- 4.) Verify the message log contains the following error: *Gentry I/O backup contactor error.*
- 5.) Select FILE —> EXIT.
- 6.) Select DISMISS.
- 7.) Select CLEANUP.
- 8.) Wait 90 seconds for the message "Scanning Hardware Reset Successful" to appear on the screen.

AXIAL ENABLE SWITCH TEST

- 1.) Turn OFF axial drive enable switch S2 on the STC backplane.
- Note: For the initial condition, do not leave the tube at the 2:30 position.
- 2.) Clear the gantry are for rotation.
 - 3.) Press the alignment light push button.
 - 4.) Verify that the gantry did not rotate.
 - 5.) Turn ON axial drive enable switch S2 on the STC backplane.

ROTATION SAFETY CHECKLIST

- 1.) Turn OFF enable switch S2 on the gantry.
- 2.) Turn OFF HV enable switch S3.
- 3.) Press red E-STOP button.
- 4.) Manually rotate the gantry 360 degrees. (Keep one finger on the Gantry button.)
 - Listen for any interference between the rotating and stationary parts. (Correct any interference problems.)
 - Listen for any loose parts. (Tighten parts as needed.)
- 5.) Turn ON the enable switch S2 on the gantry.
- 6.) Turn ON the Drive power switch S1.



WARNING

MAKE SURE THERE ARE NO OBSTRUCTIONS AROUND THE GANTRY. PRESSING THE ALIGNMENT LIGHT PUSHBUTTON WILL CAUSE THE GANTRY TO ROTATE.

- 7.) Press the alignment light push button.
- 8.) Verify that the gantry rotates.
- 9.) Perform a 4 second X-ray OFF scan.



NOTICE

During the scan, it may be necessary to enter the scan room, to obtain a better listening position. If so, keep a finger on one of the four E-STOP buttons (on the gantry), to quickly stop the gantry, if necessary.



- a.) From the console, click on the SERVICE DESKTOP icon.
- b.) Select DIAGNOSTICS.
- c.) Select DIAGNOSTIC DATA COLLECTION
- d.) Set the scan time to 4.00 seconds
- e.) Leave the door open. (This makes it easier to hear any loose or interfering parts.)
 - * Listen for any interference between the rotating and stationary parts.
(Correct any interference problems.)
 - * Listen for any loose parts.
(Tighten parts as needed.)
- 10.) After completing the 4 second scan, repeat steps 9a through 9e, with the following scan times:
 - 2.0 second scans
 - 1.0 second scans
 - 0.7 second scans
 - 0.5 second scans

1.6 Emergency Stop Check

- 1.) Use the gantry push-buttons to advance the cradle about 0.5m (2ft) from the home position.
- 2.) Press one of the E-STOP buttons on the gantry.
- 3.) Make sure the TABLE POWER shuts off, and the green LED flashes.
- 4.) Depress one of the table elevation buttons, to verify the emergency stop disabled table elevation.
- 5.) Depress one of the cradle drive buttons, to verify the emergency stop disabled the cradle drive.
- 6.) Press one of the RESET buttons to turn on X-RAY DRIVES POWER. (120 VAC LED stops flashing.)
- 7.) Press the other E-STOP button on the gantry.
 - a.) Make sure the Table Power shuts off.
 - b.) Manually move the cradle to the home position to make sure the cradle clutch released.
 - c.) Make sure the cradle latches securely in the home position.
- 8.) Press one of the RESET buttons to turn on X-RAY DRIVES POWER.
- 9.) Press one of the four table tape switches to make sure the X-RAY DRIVES POWER shuts off.
 - a.) Press one of the RESET buttons to turn on X-RAY DRIVES POWER.
 - b.) Repeat with the three remaining table tape switches.
- 10.) Press the console emergency stop switch; make sure the Table Power shuts off.
- 11.) Press one of the RESET buttons to turn on X-RAY DRIVES POWER.

- 12.) Exit.
- 13.) Return to the Service Desktop home.
- 14.) Reset buttons on Gantry and STC Service Switch bank.

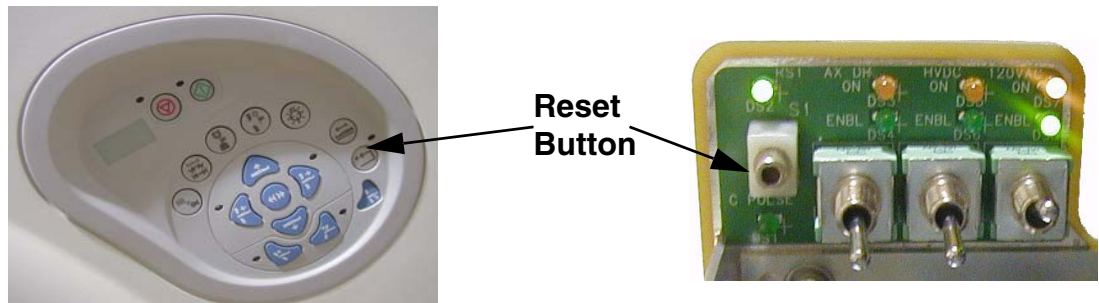


Figure 2-5 Reset buttons on Gantry and STC Service Switch bank.

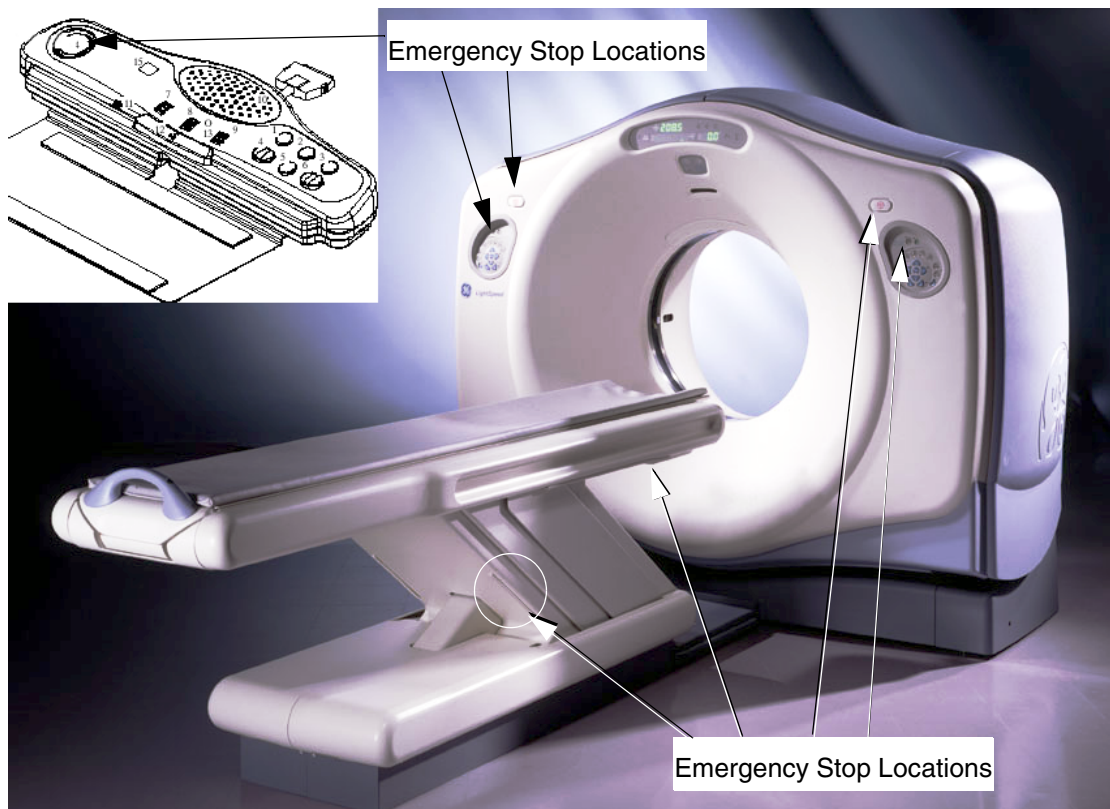


Figure 2-6 Emergency Stop button positions

Note: Emergency Stop buttons are located on the front and rear of the gantry (8 in all). They are also located on both sides of the table base (4 in all) as noted in [Figure 2-6](#).

Section 2.0 Computer Integration

2.1 Introduction

This Section describes the reconfiguration, system state restore, options, and monitor adjustment procedures.

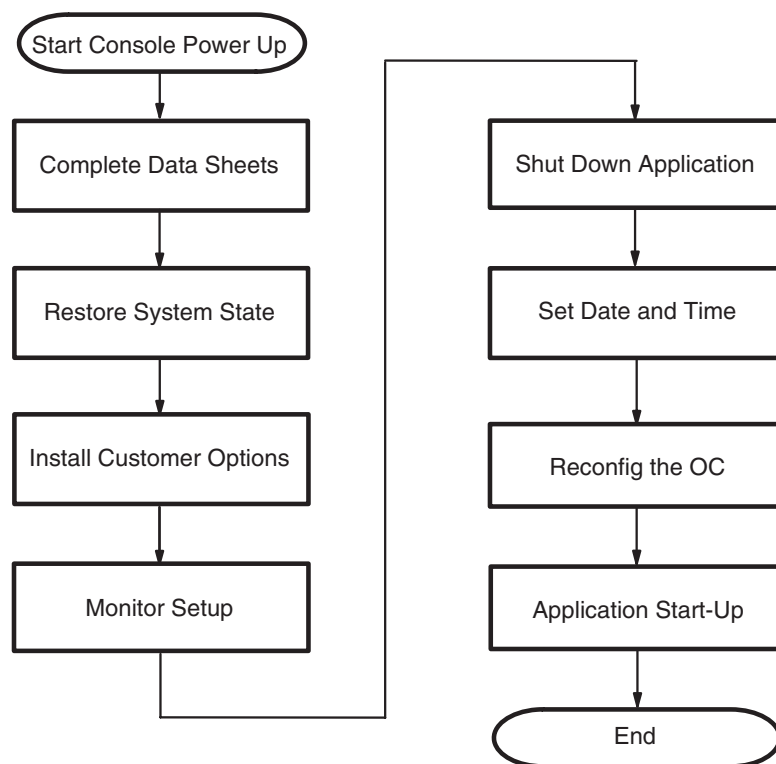


Figure 2-7 Computer Integration Process Overview

2.2 System Configuration Data Sheets

2.2.1 Requirement

Record valuable system information in the data sheets that follow. Consult with your customer or network administrator to obtain the information. Understanding how the customer plans to use their LightSpeed 4.X scanner, and their network and filming expectation reduces the time required to reconfigure the system.

- [Table 2-3 Manual Film Composer Options, on page 24](#)
- [Table 2-4 System Network Configuration, on page 25](#)
- [Table 2-5 Networking Application \(Image transfer\) Configuration, on page 25](#)
- [Table 2-6 DASM Laser Camera Configuration, on page 26](#)
- [Table 2-7 DICOM Print Camera Configuration, on page 27](#)
- [Table 2-8 DICOM Print Camera Advanced Configuration, on page 27](#)

2.2.2 Manual Film Composer Options

MANUAL FILM COMPOSER OPTIONS	
Slide Format (if available):	
Greyscale:	
Auto Printing:	
Auto Clear Page:	
Icon Labels:	
Expose Order:	
No. of Copies:	

Table 2-3 Manual Film Composer Options

2.2.3 System Network Configuration

SYSTEM NETWORK CONFIGURATION			
	FIELD NAME:	SETENV NAME:	FIELD VALUE:
System Settings:	Service ID	SERVICE_ID	
	Hospital Name	HOSPITAL_NAME	
	Exam Number *	* ask customer or checklog	
	DAS Type	DASTYPE	
	PDU Type	PDUTYPE	
Network Settings:	Gateway Host Name	GATEWAY_HOSTNAME	
	Gateway IP	GATEWAY_IP	
	Gateway Net Mask	GATEWAY_NETMASK	
	Gateway Broadcast Mask	GATEWAY_BROADCAST	
	Suite Name	SUITEID	
Option	Network Printer IP Address		
Option	HIS Server IP Address		
Option	HIS Server AE Title		
Option	HIS server AE Port		
Option	CT Server AE Title		
Option	Connect Pro IP Address		

Table 2-4 System Network Configuration

2.2.4 Network Application (Image Transfer) Configuration

Record the network application (image transfer) configuration.

NETWORKING APPLICATION (IMAGE TRANSFER) CONFIGURATION				
AE TITLE OR HOST NAME	NETWORK ADDRESS	NETWORK PROTOCOL	PORT NUMBER	COMMENTS

Table 2-5 Networking Application (Image transfer) Configuration

2.2.5 Host Ethernet Address

_____:_____:_____:_____:_____:_____

2.2.6 Camera Application Configuration

Record the camera application configuration for the DASM or DICOM print camera.

DASM LASER CAMERA CONFIGURATION	
Camera Type:	
DASM Type:	
Film Smooth/Sharp Setting:	
Options:	
Valid Film Formats:	
Default Film Formats:	

Table 2-6 DASM Laser Camera Configuration

DICOM PRINT CAMERA CONFIGURATION	
Camera Type:	
Host Name:	
IP Address:	
AE Title:	
TCP/IP Listen Port:	
Comments (Optional):	
Valid Film Formats:	
Default Film Formats:	
Destination:	
Orientation:	
Medium Type:	
Magnification Type:	

Table 2-7 DICOM Print Camera Configuration

DICOM PRINT CAMERA ADVANCE CONFIGURATION	
Smoothing Type:	
Configuration:	
Minimum Density:	
Maximum Density:	
Empty Density:	
Border Density:	
Association Timeout:	
Session Timeout:	
N-Set Timeout:	
N-Action Timeout:	
N-Create Timeout:	
N-Delete Timeout:	
N-Get Timeout:	

Table 2-8 DICOM Print Camera Advanced Configuration

2.3 Configuration

Note: Type the text shown in **boldface**, and press the ENTER key on the keyboard.

2.3.1 Restore System State

Tools Required: None

Your system should have a system state disk (DVD, for Linux, or MOD, for IRIX), located in the software collector box. The system state disk contains:

- Collimator Characterization
- Phantom Calibrations
- Gen Cal
- Other Data

The installation process uses all the system state files. At this time, use the system state DVD (or MOD) to restore all files.

If you cannot locate an existing system state disk, you must recalibrate your system.

- 1.) Insert the System State DVD into the DVD drive (or MOD into MOD drive), in the SCSI Tower.

- 2.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



Comment: If the Service Key is installed, the user has to press the (1), (2), (3) keys in order to proceed.

- 3.) Click on the PM icon.
- 4.) Select SYSTEM STATE.
- 5.) Select CHARACT.
- 6.) Select CALS.
- 7.) Select RESTORE to restore the system characterization and phantom calibration files to the system.



Note: **Restore State can take as long as ten minutes, although typical times average about three minutes. When Restore State completes, dismiss the tool, and proceed to the next section.**

If any error should occur during the restore process, see the *Software Installation Procedures* manual (Load From Cold) for information regarding possible error messages and their recovery.

- 8.) Click NO for Reset Scan Hardware popup message.
- 9.) Select DISMISS.

2.3.2 Install Customer Options

Tools Required: None

2.3.2.1 Software

Note: **Your system may have media containing customer purchased options. If your system has an options disk, install it at this time—otherwise skip this section.**

Ensure that the options disk (DVD or MOD) is NOT write protected at this time. The initial install requires that the DVD/MOD be write enabled; subsequent installs can be done with the DVD/MOD write protected.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.



- 3.) Select INSTALL.

- 4.) Select INSTALL OPTIONS.

An Options Window appears.

- Check the FDO to see what options were ordered.
- Compare FDO options to those on the Option DVD/MOD.
- If different contact your local sales representative.

2.3.2.2 Camera

Tools Required:

- Small flat blade screw driver
- Data collected from data sheets (see See “Camera Application Configuration” on page 26.)
- *Software Installation Procedures* manual
- *System Service Manual*.

Note: If a DASM is requires, the DASM hardware must be installed before proceeding.
For details on camera configuration, refer to the *Software Installation Procedures* manual.
For details on troubleshooting the camera, refer to the *System Service Manual*.

- 1.) Click on the SERVICE DESKTOP icon.



- 2.) Select UTILITIES icon.



- 3.) Select INSTALL CAMERA.

- 4.) Select ADD.

- 5.) Select DASM or DICOM.

- 6.) Follow procedures on the screen.

- 7.) Return to Home Page

2.3.3 Monitor Setup

This section provides the basic information and procedures needed to set up and adjust the monitors on the LightSpeed 4.X. Two models of monitors are being used, and the system you are working on may be equipped with either one. Check the model number on the rear of the monitor to determine which of the following procedures to use.

Tools Required: Sony Service Manual (located in monitor box)

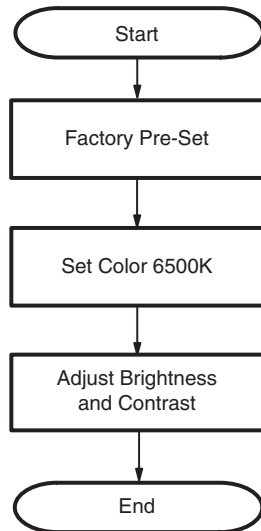


Figure 2-8 Monitor Setup Flowchart

2.3.3.1 Sony Model GMD-5402 -- GE #2254212

Using Menu Buttons

- 1.) Press the Main Menu button to display the main menu on your screen.



Figure 2-9 Main Menu Button

- 2.) Press the  buttons to highlight the menu you want to adjust. Press the Main Menu button to select the menu item.

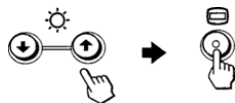


Figure 2-10 Select Menu Item

- 3.) Press the  buttons to select the desired adjustment item. Press the  buttons to make the adjustment.



Figure 2-11 Adjust the Menu

- 4.) Press the Main Menu button (Figure 2-9) once to return to the main menu, and twice to return to normal viewing. If no buttons are pressed, the menu closes automatically after about 30 seconds.
- 5.) Press the reset button to reset the adjustments back to the factory settings.



Figure 2-12 Reset Button

Color Adjustments

- 1.) Adjust the ambient room light to the level used by the operator during image review.
- 2.) Select the COLOR menu to adjust the picture's color temperature. (You can use this to match the monitor's colors to a printed picture's colors.)

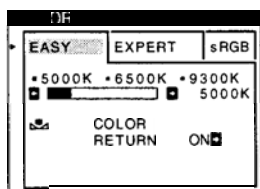


Figure 2-13 Color Menu

- 3.) Set monitor color temperature to 6500K. Refer to the system service manual for this procedure, if necessary.

Brightness and Contrast Adjustment

Brightness and contrast adjustments are made using a separate BRIGHTNESS/CONTRAST menu. These settings are stored in memory for the signals from the currently selected input connector.

- 1.) Press either one of the brightness or contrast buttons. The BRIGHTNESS/CONTRAST menu appears on the screen.

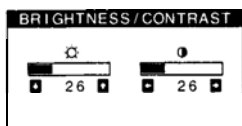


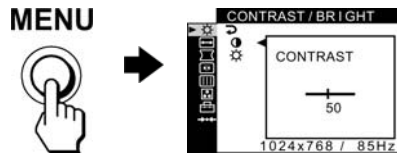
Figure 2-14 Brightness/Contrast Menu

- 2.) Press the / buttons to adjust the brightness and the / buttons to adjust the contrast.

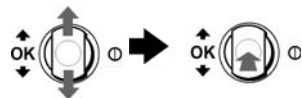
2.3.3.2 Sony Model GMD-F520 -- GE #2333243

Using Menu Buttons

- 1.) Display the menu by pressing the MENU button.



- 2.) Select the desired menu by toggling the control button up or down to highlight the menu.



- 3.) Toggle the Control button up or down to make the adjustment. Then press the Control Button.

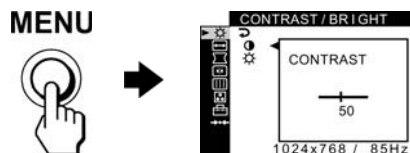


- 4.) Press the Menu button to close the menu.



Selecting the On-Screen Language

- 1.) Display the menu by pressing the MENU button.



- 2.) Toggle the control button up or down to highlight  OPTION. Then press the control button.



- 3.) Toggle the Control button up or down to highlight  LANGUAGE. Then press the Control Button.



- 4.) Toggle the Control button up or down until the desired language appears on the screen. Then press the Control button to select the language.
- 5.) Press the Menu button to close the menu.

Color Adjustment

You can use this to match the monitor's colors to a printed picture's colors.

- 1.) Adjust the ambient room light to the level used by the operator during image review.
- 2.) Press the Menu button. The menu appears.
- 3.) Toggle the Control button up or down to highlight  COLOR. Then press the Control button.
- 4.) Toggle the Control button up or down to highlight  COLOR on the adjustment bar. Then press the Control button.
- 5.) Toggle the Control button up or down to fine tune the color temperature.

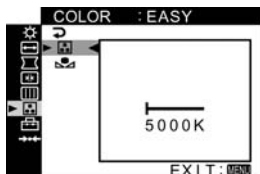


Figure 2-15 Color menu

- 6.) Set monitor color temperature to 6500K.
- 7.) Press the Menu button to exit.

Brightness and Contrast Adjustment

- 1.) Press the Menu button. The menu appears.
- 2.) Toggle the control button up or down to highlight . Then press the Control button, and the Contrast/Bright menu appears.

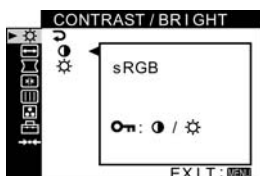
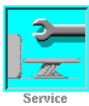


Figure 2-16 Contrast/Bright menu

- 3.) Toggle the Control button up or down to highlight  (brightness) or  (contrast). Then press the Control button.
- 4.) Toggle the Control button up or down to adjust the contrast or brightness.

2.3.4 Contrast/Brightness Adjustment Procedure

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon. 
- 2.) Click on the IMAGE QUALITY icon. 
- 3.) Select INSTALL SMPTE IMAGE and wait approximately 3-4 minutes for SMPTE image to install. When complete the following message will display:
SMPTE and QA images have been successfully copied
- 4.) Press ENTER to exit the Service Desktop.



- 5.) Click the IMAGEWORKS icon.
 - 6.) Display the SMPTE pattern. Use the browser to select Exam 1000, which contains the SMPTE pattern, and enlarge the image to full screen display.
 - 7.) Select Viewer.
 - 8.) Select 1:1 format.
 - 9.) Increase the monitor's contrast to maximum.
- Note: Adjust monitor contrast until the operator sees the anatomical structure (window raster)
- 10.) Increase the Brightness to maximum.
 - 11.) Decrease the Brightness, until the raster just fades into, and matches, the monitor screen background. At this point, the 5% and 95% patches should be just visible.
 - If additional tweaking is required to attempt to match the monitor image to the filmed image, use only the brightness control.
 - If the CRT image exhibits any tearing or smearing of the alphanumeric characters, then reduce the contrast setting slightly until the tearing/smearing is just eliminated. The optimum setting for contrast is the highest setting that does not cause tearing/smearing of the alphanumeric characters.
 - You should always finish up by displaying and filming images of anatomy (typical heads and bodies), and asking the technologist to compare the CRT image to the film image.
 - 12.) To set up the text (left) monitor:
 - Swap the right and left monitor's video cables to view the SMPTE pattern on the left monitor.
 - Repeat steps 2-12

2.3.5 Shut Down Application

Tools Required: None

Standard Level

If Applications is currently running, you must shutdown system applications.



- 1.) Click on the SERVICE DESKTOP icon.
- 2.) On the desktop toolbar select UTILITIES icon.
- 3.) Select APPLICATIONS SHUTDOWN (to bring down applications only).



Super User Level

- 1.) Open a Shell window.
- 2.) Type: `su` - **ENTER** at the prompt.
- 3.) Type the root (super user) password: `#bigguy`

2.3.6 Check/Set Date and Time

If date and time need to be corrected:

- 1.) Type/enter `setdate` ENTER.
- 2.) Follow the instructions to set the correct time and date.

2.3.7 Reconfig the OC

Note: The document collector box that arrived with your system contains a *Software Installation Procedures* manual, which documents the reconfiguration procedure in more detail.

2.3.7.1 Overview

On the following screens, make the changes necessary, pressing the corresponding button at the top of the screen to move from screen to screen. When done, either press the ACCEPT button to start the reconfiguration process, or press QUIT, to exit without changing the system configuration. While the reconfiguration is going on, messages are displayed in a shell window that closes when reconfiguration is complete. The reconfiguration output is logged to the following file:

```
/var/adm/install.log.YYYYMMDDWWHHMMSS
```

Where

YYYYMMDDWWHHMMSS is the Date/Time that the reconfiguration was started.

To view the file, type: `more /var/adm/install.log.YYYYMMDDWWHHMMSS`

It is possible to abort the reconfiguration while entering information on the reconfiguration screens. Simply press the QUIT button at the top of the screen. There is NO safe way to abort the reconfiguration after pressing the ACCEPT button. If the entries made in the screens were incorrect, DO NOT try to stop the reconfiguration, instead wait for it to complete, and rerun reconfig, entering the correct parameters.

2.3.7.2 Procedure

- 1.) Change directory to scripts:
Type: `cd /usr/g/scripts` ENTER at the prompt.
- 2.) Launch the Install utility:
Type: `reconfig` ENTER at the prompt.

The OC displays the Install Utility Window as shown in [Figure 2-17](#).

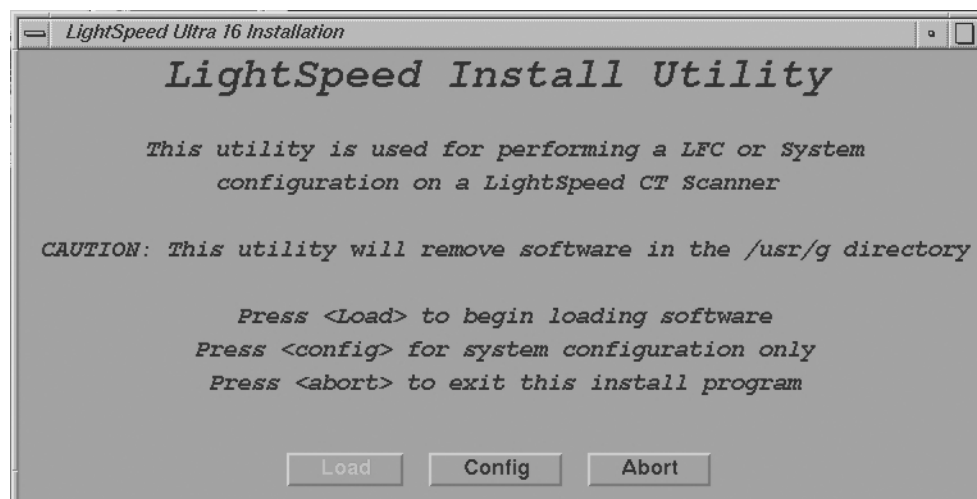


Figure 2-17 Install Utility Window

Comment: The following pages show the screens that are used to change the configuration of the system. These screens are the same as those used for the Software Configuration during Load From Cold. The actual screens will vary depending on the current configuration of your system.

- 3.) Click on the CONFIG button.

The OC displays the System Configuration - System Settings screen, as shown in [Figure 2-18](#).

Figure 2-18 System Settings Screen

- 4.) Configure System Settings

- a.) Hospital Name ([Figure 2-18](#), item a) configures the name that will show up on images produced by this scanner. *Example:* ST MARYS HOSPITAL
- b.) Service ID ([Figure 2-18](#), item b) is issued by the Service organization. *Example:* 262785CT2 (no spaces)
- c.) Select the Time Zone for the site.
 - * Use the scrollbar at the bottom of the time-zone selection list to view the entire description of the time-zone you are about to select, to ensure that you are selecting the correct time-zone for your location.
 - * If the time-zone of your location is listed, select one of the universal times instead. In this case, automatic changes for daylight savings time will not take effect. Refer to the *Software Installation Procedures* manual, if additional information regarding time-zone setting & selection is required.
- d.) Next MOD # *Not Currently Implemented.*
- e.) Next Patient Exam # configures the next Exam number the scan user interface will use. At initial system installation, type: 1
- f.) Next Diagnostic Exam # *Not Currently Implemented.*

- g.) Regenerate Database determines whether the Scan Database will be regenerated during reconfiguration. Select NO.

Note: If YES is selected, then all scan data will be deleted on the Scan Data Disk. To avoid accidental regen, NO is selected by default each time reconfig is run. Regenerating the Scan Database destroys all the scan and calibration files on the Scan Data Disk. Make sure that all images have been reconstructed before regenerating the Scan Database.

This procedure is normally only needed when a Scan Data Disk is replaced or reformatted. Refer to the *Software Installation Procedures* manual, if you require more information regarding Scan Database Regeneration.

- h.) Mobile System Select to tell the software if this CT is in a mobile environment or not.

- 5.) Select the PREFERENCES button to display the Preference Settings screen (Figure 2-19).

Figure 2-19 Preferences Setup Screen

- 6.) Configure Preferences Settings
- Doctors Title - Enter the title for the Doctor. (eg. radiologist)
 - Units for Patient Weight - Tells the software whether pounds or kilograms are being used.
 - Language configures the language to be displayed on the Application screens (ENGLISH, GERMAN, PORTUGUESE, FRENCH, ITALIAN, SPANISH, JAPANESE).
 - Preferred Fast Cal KV configures the preferred kV that the Fast Cal Routine will calibrate (80, 100, 120, 140 in the Selected Preferred Fast Cal KV field). The default selections are 120 and 140.
- Comment:* These kVs should include all kVs that the site uses for patient scanning. Deselecting All Preferred FastCal KVs is the same as selecting ALL the Preferred FastCal KVs
- Date Format configures the format in which the date will be displayed on the images.
 - Time Format configures the format in which the time will be displayed on the images

- g.) Default Archive Device is DICOM.
 - h.) HIPAA Present - Consult with the customer. Unless they specify that they want to use this feature, this field should be set to OFF.
 - i.) Select the site preferred Dose Information Display option for the site to use in monitoring calculated Patient Dose:
 - * Select ON (full CTDiw Display)
 - * Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or
 - * Select OFF (no CTDiw Display)
- 7.) Select the HARDWARE button to display the Hardware Settings Screen (Figure 2-20).

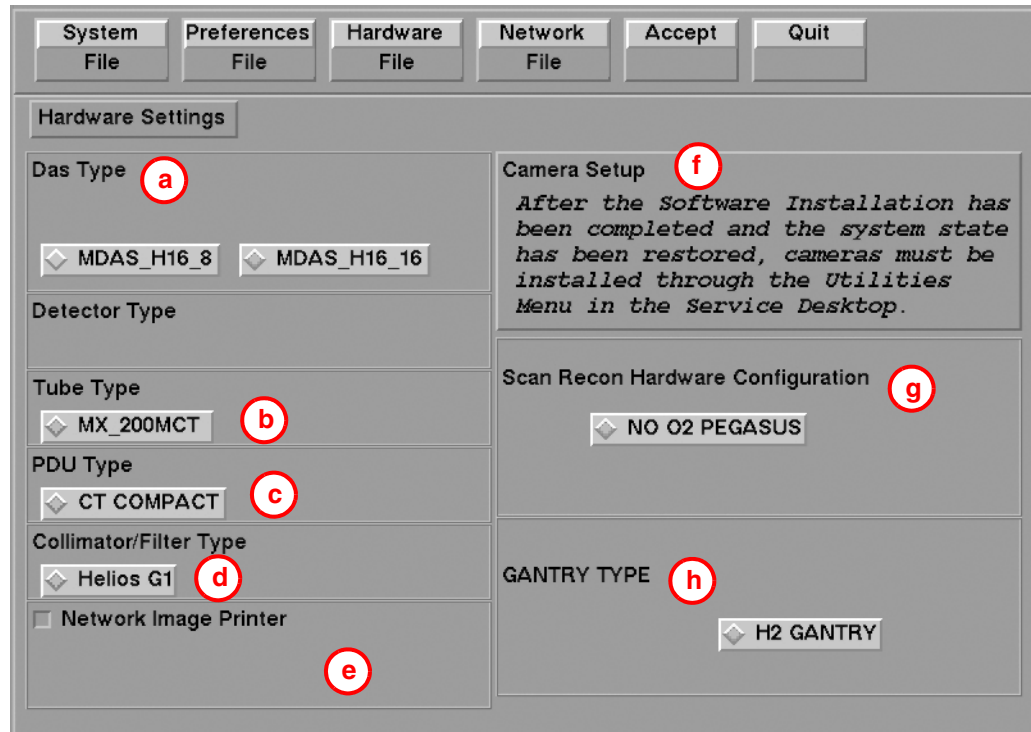


Figure 2-20 Hardware Settings Screen

- 8.) Configure Hardware Settings
- a.) DAS Type configures which DAS the software should expect.
 - b.) Tube Type configures which X-Ray Tube the software should expect.
 - c.) PDU Type configures which PDU the software should expect.
 - d.) Collimator/Filter Type configures which Collimator/Filter the software should expect. Presently, LightSpeed 4.X systems only support HELIOS G1.
 - e.) Select Network Image Printer only if the site plans to use a network connected postscript printer (such as Codonics printers) with this system.

Note: A "PRINT SERVER" IS NOT SUPPORTED (i.e., a computer with a printer attached); a Network Image Printer (which is supported) is a printer attached directly to the network.

- f.) Camera Setup:
- Comment:* Camera Setup is not performed using Reconfig. It is performed from the Service Desktop, once Applications have been started. See "DICOM Filming Devices Setup" on page 85.
- g.) Scan Recon Hardware Configuration selects whether the O2 is present.

- h.) Gantry Type selects the type of Gantry connected to this console
- 9.) Select the NETWORK button to display the Network Settings Screen as shown in [Figure 2-21](#).

Figure 2-21 Network Settings Screen

- 10.) Configure Network Settings

Comment:

This screen provides the ability to declare the CT system on a hospital network. Key information such as Host Name, IP Address, Net Mask (for CT systems on a subnet) must be obtained from the hospital network administrator.

See Chapter 4 for more information and complete details of setting the Hospital/System Network Configuration.

- a.) Enter the Suite Name.

The Suite Name must start with a letter, followed by three alphanumeric characters. Total must be four characters long. The name of the OC interface will be <Suite Name>, within the scanner's subnet.

Typically, you should use su01 or ct01 ("su" and "ct" must be lowercase), unless the customer prefers a different suite name.

- b.) Enter the hospital provided Host Name.

The Host Name identifies the network hostname and AE Title of the CT system to the hospital's network. The Host Name:

- * **MUST NOT** exceed 16 Characters.
- * **MUST** only contain the following characters: A through Z, a through z, 0 through 9, - and _.

Comment:

The Host Name is typically stmary or ct01.

- c.) Enter the hospital provided IP Address for the system.
- d.) Enter the hospital provided Net Mask (if the CT system is on a subnet).
- e.) Enter the Broadcast Address.

Comment: The Broadcast Address should be the same as the IP Address except for the bits of the host id portion (last digit group) set to 1's or 0's depending on the configuration of the network. The standard default is 1's, but older Sun OS machines used 0's.

For example:

If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the Broadcast Address.

If the network contains genesis based scanners or other Sun OS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.

- f.) Enter the hospital provided Default Gateway IP Address in the Default Gateway field (if applicable). If the site network does not use a default gateway, leave the field blank.
- g.) Select NIS (a.k.a. Yellow Pages database) Advanced Option—only if requested by the hospital network administrator—as follows:
 - * Select ADVANCED OPTIONS button on the Network Settings Screen.
 - * Select USE NIS button.
 - * Enter the hospital provided Domain Name (from the hospital network administrator).
 - * Review all pages to be sure the information entered is correct before proceeding to the next step.

- 11.) Select ACCEPT on the System Configuration Screen.

Comment: The system loads the CT Application Software, OS patches, kernel changes and configures the system on the OC.

The loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a shell window that closes when the loading process is complete. All the window output is logged to a file named:_____

- 12.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click on YES. (See [Figure 2-22](#).)



Figure 2-22 Reboot Screen

- 13.) The system will automatically login as `ctuser` after the reboot. Select OK on the Autostart Disabled popup message.

2.3.8 Applications Start-Up

Open the Console shell window, and type: `st` ENTER. The applications desktop appears on the monitor.

Section 3.0

Table Gantry Integration

3.1 Introduction

Use these procedures to functionally check every part of the table/gantry subsystem.

REQUIRED TOOLS

- Multimeter

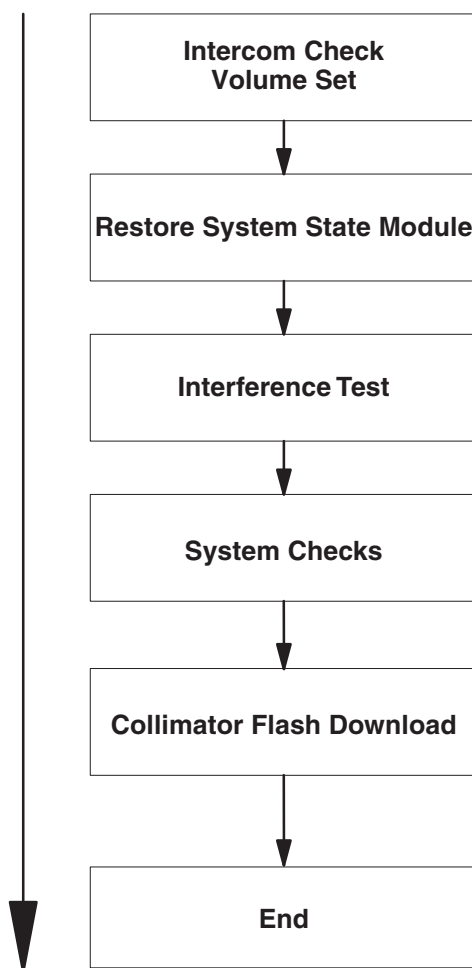


Figure 2-23 Table/Gantry Integration Flow Chart

3.2 Autovoice/Intercom Check

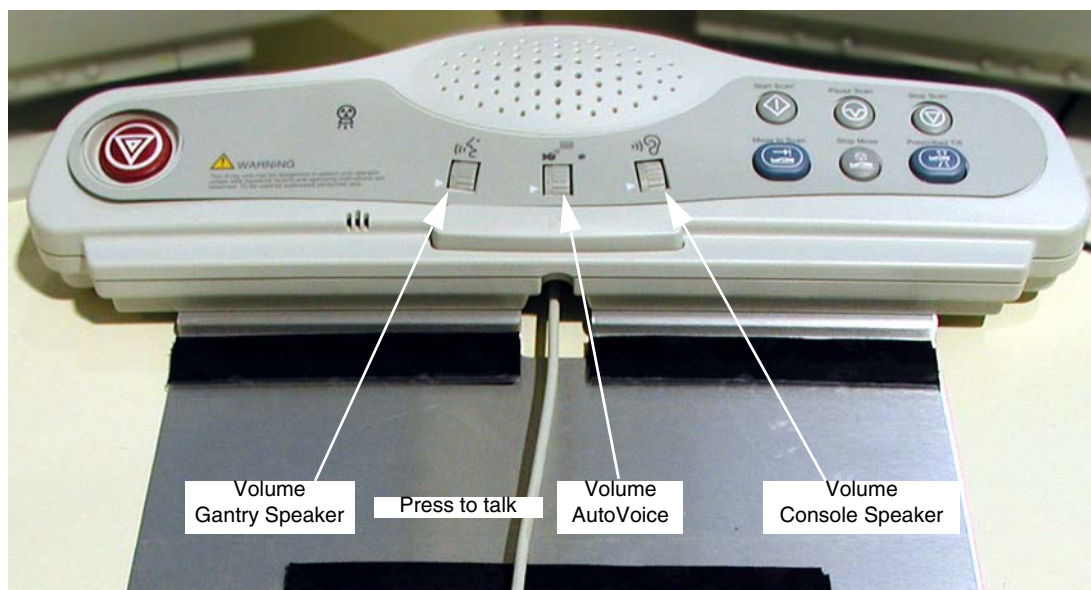


Figure 2-24 SCIM Volume Controls

3.2.1 Gantry Speaker

To adjust the gantry speaker volume, adjust the left-most volume thumb wheel while speaking into the console microphone (press the bar on the SCIM).

3.2.2 Console Speaker

To adjust the console speaker volume:

- 1.) Click on the Autovoice volume control from the tool chest.
- 2.) Click on the Audio Panel GUI icon to open it.
- 3.) Leave all audio panel sliders at their default positions, as shown in [Figure 2-25](#).

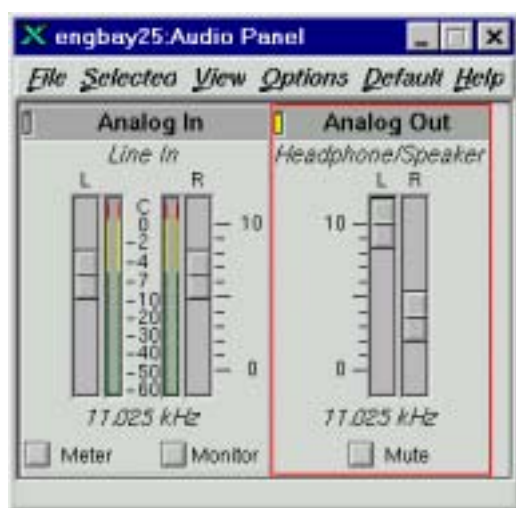


Figure 2-25 Audio Panel

- 4.) Select FILE-SAVE when finished to retain your settings.

3.2.3 Autovoice Volume

- 1.) On the Scan Desktop, select PROTOCOL MANAGEMENT.
- 2.) Select AUTO VOICE RECORD.
- 3.) Click on the 3.4 button, to the right of "FF2. Inspiration".
- 4.) Click on the PLAY button, to play the Inspiration AutoVoice message.
- 5.) Adjust the center volume thumb wheel, while Autovoice is playing, to set the volume for the gantry speaker.
- 6.) Repeat steps 4 and 5, as necessary to achieve satisfactory volume.
- 7.) Select DONE, then Select QUIT.

Note: If a satisfactory volume can not be achieved, refer to the *LightSpeed 4.X System Service Manual* and review the factory setup for the:

- console intercom pots
- gantry intercom pots

3.3 Table/Gantry Characterizations

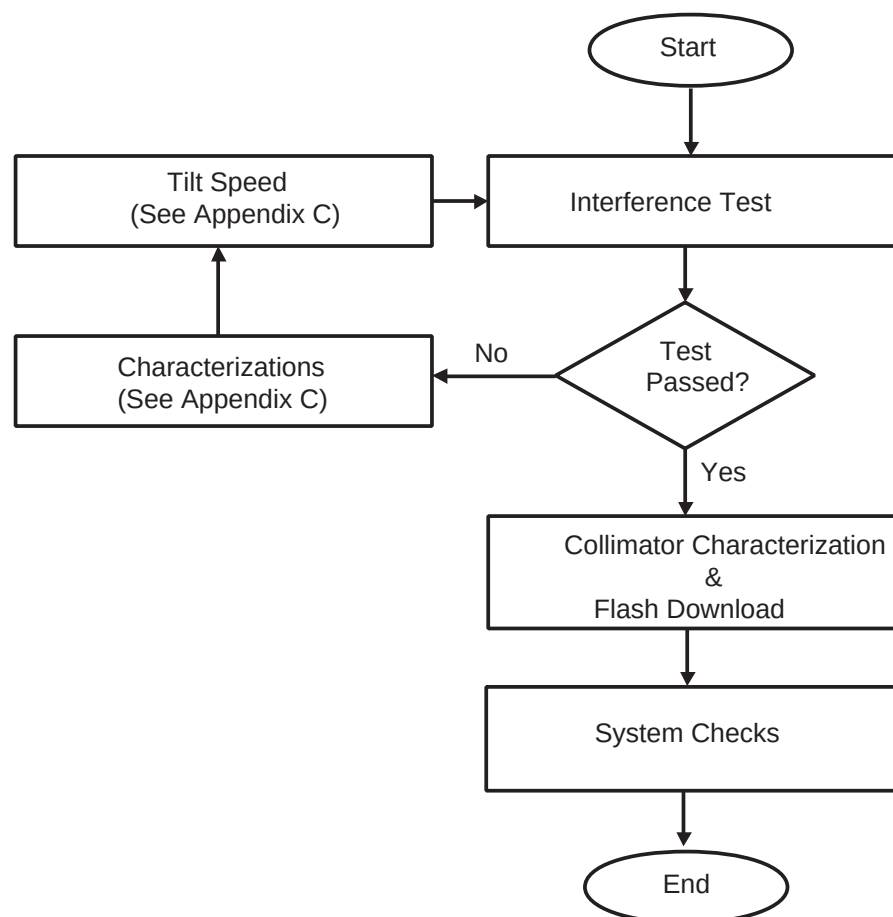


Figure 2-26 Characterization Flowchart

3.4 Interference Test

Tools Required: None

Note: Be sure that the System State was restored from **MOD** per Section 2.0.

Reset the hardware to download the new characterization values before proceeding to the table/gantry interference test on this page.

If the following elevation and/or tilt display readings fail to meet specifications, DO NOT adjust the limit switches. Instead, re-characterize and/or adjust tilt speed. Refer to Appendix C. Perform elevation and cradle first, then repeat the test. If it still fails, perform tilt.



NOTICE **Interference Test Requirement: No tilt motion shall cause the gantry tilting frame to touch the stationary base covers for any tilt angle.**

1.) Select SYSTEM RESETS.

Press only the **fast** push-button to display the elevation limits for the present elevation/tilt combination.

- a.) Move the cradle to the ISO center (maximum height) position.
- b.) Use the DOWN pushbutton to lower the table to the minimum height.
- c.) Make sure the elevation display reads 500.0 ± 3 mm.
- d.) Use the UP pushbutton to raise the table to the maximum height.
- e.) Make sure the elevation display reads 25.0 ± 3 mm.

Note: If this step fails, check table/gantry level (Run table elevation characterization if necessary).

“I” tilt means the top of the gantry tilts away from the table. “S” tilt means the top of the gantry tilts toward the table.

2.) Advance the cradle about 36 inches from the home position.

- a.) Tilt the gantry to $I23$ degrees.
- b.) Use the DOWN pushbutton to lower the table until it automatically stops.
- c.) Make sure the elevation display reads 115.5 ± 3 mm.
- d.) Make sure the table clears the gantry covers by at least $\frac{1}{2}$ inch.

3.) Tilt the gantry to $I20$ degrees.

- a.) Use the DOWN pushbutton to lower the table until it automatically stops.
- b.) Make sure the elevation display reads 210.0 ± 3 mm.
- c.) Make sure the table clears the gantry covers by at least $\frac{1}{2}$ inch.

4.) Use the UP push-button to raise the table until the elevation display reads between 115.5mm and 210mm.

- a.) Tilt the gantry to $I10$ degrees.
- b.) Tilt the gantry away from the table; make sure it stops at $I22.0 \pm 0$ degrees.
- c.) Tilt the gantry toward the table; make sure it stops at $S18.5 \pm 0$ degrees.

5.) Return the cradle to home position.

- a.) Make sure you can tilt the gantry to 30.0 ± 0 degrees.

Note: The Gantry tilts at 1 degree per second. If the speed is less than this, see “Tilt Speed Adjustment Procedure” on page 118.

- b.) Leave gantry tilted to $I30$ degrees for now.

6.) Raise the table until you can move the cradle.

- a.) Advance the cradle about 12 inches from the home position.

- b.) Use the **HOME** Button to return the cradle to the home position, tilt the gantry to 0 degrees, and lower the table to minimum height.
Make sure you cannot move the cradle at table heights below 210.0.
- c.) Verify that the *cradle lock* indicator on the gantry display is lit.

3.5 Check Alignment Lights

3.5.1 Turn Alignment Lights ON



CAUTION Verify all personnel have cleared the system. The Gantry rotates during this check.



- 1.) Turn ON the Axial Drive Enable and HVDC Enable switches (located on the STC backplane – refer to [Figure 2-4, on page 20](#)).
- 2.) Press the alignment light button on the gantry control panel.
The gantry will rotate and the lights will turn ON.



**WARNING
LASER EYE
INJURY**

NEVER STARE DIRECTLY INTO THE LASER BEAMS WHEN YOU OPERATE THE ALIGNMENT LIGHTS. STARING INTO THE BEAMS CAN CAUSE PERMANENT EYE DAMAGE.

3.5.2 Internal Axial Lights

- 1.) Place a sheet of plain white paper over the output port of each light.
- 2.) Verify that the two laser lines coincide and appear as a single line.

Note: **GE designed the internal axial lasers on the current CT system to shine *down* on the collimator. Do NOT adjust the internal alignment lights at this time. The tomographic plane tests use the QA phantom to check the internal axial lasers alignment to the collimator.**

3.5.3 External Axial to Internal Axial Distance

Note: **Ensure that cradle is level.**

- 1.) Raise the table to its highest elevation.
- 2.) Extend the cradle until you see both the internal and external laser lights shining on the cradle.
- 3.) Place a metric rule on the right edge of the cradle, and measure the distance from the internal axial laser line to the external axial line. Verify this distance equals 240.0 mm $\pm$ 1.0 mm.
- 4.) Place the rule on the left edge of the cradle, and measure again.
- 5.) Leave the cradle in its current position, and lower the table to the minimum elevation.
- 6.) Measure the distance between the internal and external lights on both edges of the cradle, as above. Verify the distance remains equal to 240.0 mm $\pm$ 1.0 mm.

3.5.4 Coronal Lights

- 1.) Place a sheet of plain white paper at the left side of the patient opening, in front of the coronal laser light. Verify the two coronal lines coincide.
- 2.) Move the paper to the right side of the patient opening. Verify the two coronal lines coincide.
- 3.) Place the paper in the center of the gantry opening. Use a level to verify that the coronal lines are horizontal.

3.5.5 Turn Lights OFF

- 1.) Press the alignment light button on the gantry control panel, again, to turn the lights off.
- 2.) Turn OFF the axial drive enable and HVDC enable switches.

3.6 Collimator Characterization & Flash Download

The Collimator Flash Download diagnostic compares the values in flash memory on the collimator control board with the values stored on the system, and updates if necessary. If the memory values don't match, the diagnostic prompts you to type/enter the collimator serial number to complete the update.

To prevent invalid test results, remove any obstructions from the gantry opening.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon, .

- 2.) Click on the UTILITIES icon, .
- 3.) Select INSTALL.
- 4.) Select FLASH DOWNLOAD TOOL.
- 5.) Select UPDATE.
- 6.) Look for "Download Successful" message.
- 7.) Select DISMISS when finished.

Note: New installations upload values from FLASH.

Chapter 3

Image Quality



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 5](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Introduction

This chapter contains descriptions of the procedures used to test system operation after installation, alignment and calibration, but prior to customer turnover and acceptance. The Image Series sections instruct you to scan system phantoms, record the means and standard deviations, and evaluate the image quality of the resulting images.

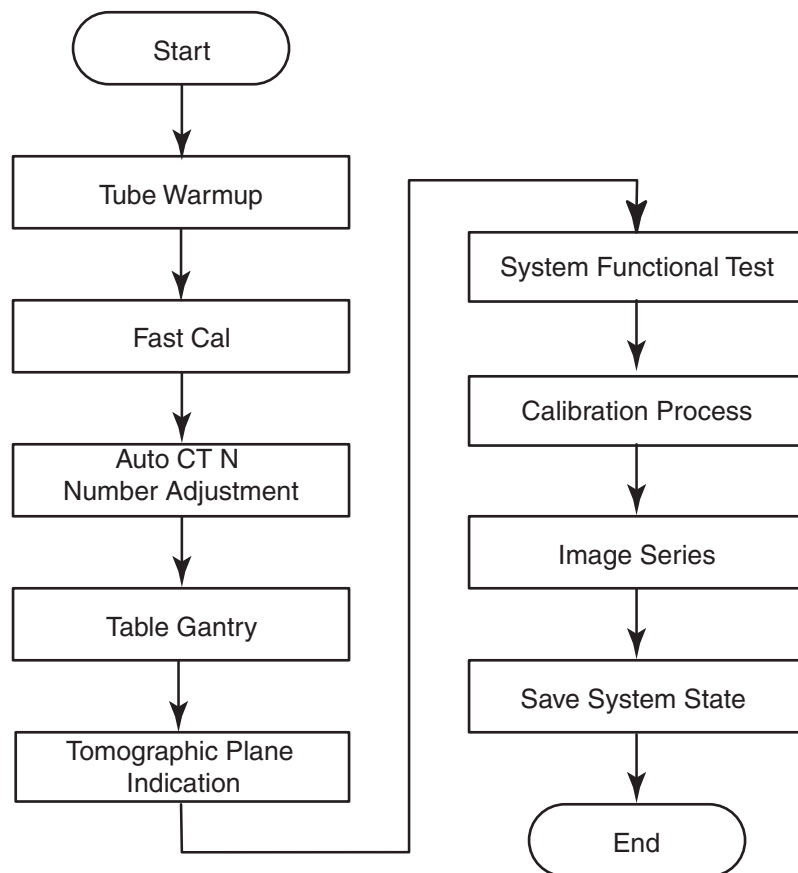


Figure 3-1 System Test Overview

Section 2.0 Calibration Process

2.1 Reference Procedures

Do NOT perform these procedures now. They are placed here as Reference for later procedures. You will be directed to follow these instructions at several points throughout this chapter.

2.1.1 Scanning with Service Protocols

Locate the Manufacturing and Installation protocols under Infant area 20.

Note: **Manufacturing and Service share this Protocol list. Different product option offerings also use this list. Carefully follow the scan section instructions, and verify you acquired the images with the correct technique before filling out the data sheet. Otherwise you may troubleshoot an image problem that only exists because you used the wrong technique.**

- 1.) Select the NEW PATIENT icon on the left monitor
- 2.) Enter a Patient ID (e.g., getest).
- 3.) Click on the box labeled PEDIATRIC.
- 4.) Select a service protocol from the list, to display the corresponding view edit screen.
Optional Method: Enter the protocol number into the Protocol Number Field on the Exam Rx Screen.

2.1.2 Center Phantom

- 1.) Select SCANNER UTILITIES icon on the left monitor
- 2.) Select CENTER PHANTOM.
- 3.) Follow the on-screen procedures.
- 4.) The phantom center spec is $\pm 5\text{mm}$.
- 5.) Select QUIT, when the phantom is within specification.
- 6.) Level the phantom both front to back and side to side. (use a 6" level)

2.2 Prepare the QA Phantom

Note: **The QA phantom is shipped water filled.**

- 1.) Locate the multi-language sticker packet in the QA phantom shipping box.
- 2.) Attach the sticker with the customer's language to the face of the phantom hanger bracket.

2.3 Calibration Process Introduction

If your system has a factory supplied state MOD, you used it to load the system calibration files during the Restore System State ([2.3.1 on page 28](#) of this manual).

Section 3.0

Table/Gantry Alignment Procedure

3.1 Conditions

This procedure may require the removal of the table bottom, side and rail covers.

This procedure requires that mechanical final alignment was completed and passed.

3.2 Procedure

- 1.) If this is a Non-Fixed Site, check the Van level. Correct, if necessary.
- 2.) Check Table level. Correct, if necessary.
- 3.) Drive the table to its highest elevation (phantom holder removed).
- 4.) Drive the table cradle in and out five times to seat the rollers.
- 5.) Turn on the alignment lights. Check the scan window for proper installation.
- 6.) Advance the end of the cradle (black dot on cradle) to the internal light.
- 7.) Complete [Section 2.1.1, "Scanning with Service Protocols"](#) now.
- 8.) Select the service protocol, PERPENDICULAR ALIGNMENT. Protocol should appear as shown in [Table 3-1](#).

Scan Type	Direction	kV	mA	SFOV	Thickness	Scan Time	Start Loc.	End Loc.	Algorithm
Axial	Head First	120	40	Large	5mm/1i	2sec	S0	S0	Bone
Axial	Head First	120	40	Large	5mm/1i	2sec	I1000	I1000	Bone
Scout		120	40			90°	S0	I1000	
Scout		120	40			0°	S0	I1000	

Table 3-1 Perpendicular Alignment Scan Parameters

Set Internal Landmark. The red boxes will disappear from the screen.

- 9.) Click on CONFIRM, then press the **START SCAN** button when lighted, to perform the scan.
- 10.) Use over/under multi-image display to view both images at the same time:
 - a.) Click on IMAGE WORKS
 - b.) Highlight the scans just performed and click on VIEWER.
 - c.) Click on FORMAT, and select the "over/under" view (two boxes aligned vertically).
- 11.) Use the upper image (cradle front) to verify that table is centered:
 - a.) Click on the "grid" icon.
 - b.) Click on MEASURE, and select the straight (angled) line.
 - c.) Using the mouse, position one end of the line on the vertical axis.
 - d.) Position the other end at one outside edge of the cradle, forming a perfectly horizontal line (angle = 90). Read the distance displayed on screen.
 - e.) Repeat step d to measure the distance from the vertical axis to the other edge.
- 12.) Repeat step 11 for the lower image (cradle at 1000 mm).

- 13.) If the edges are off by greater than 1 mm (front to back), perform the following:
 - a.) Loosen the table anchors at locations #5, #6 and #7.
 - b.) Move the table half the measured distance.
 - c.) Return to Step 8. Repeat the procedure until the table edges line up (± 1 mm). If needed a scout can be completed to check table horizontal presentation
Check the two scout scans
 - i.) Check the tube at zero scan for table gantry perpendicularity. Make table adjustments as needed
 - ii.) Check the tube at 90 scan for cradle level and visual irregularity with the white side molding. Molding strips should be parallel within ± 3 mm. Cradle should be level within ± 3 mm.
 - d.) Tighten the anchors and torque to 74.6 N-m (55 ft.-lb.) for fixed sites only. (75 ft.-lb.) for non-fixed systems.
 - e.) Reinstall all table parts removed to access the anchors, including covers, ground straps, elevation motor leads and the center support bar.
- 14.) Install all table and gantry covers.
- 15.) Use ERASE ALL to remove lines and grids from the images.
- 16.) Click on END EXAM to exit the program.

Initial: _____

Section 4.0

Tube Warm Up and Fast Cal



- 1.) Select  to warm up the tube.
- 2.) Select FAST CALIBRATION from the Daily Prep menu.

Note: **Use the default Fast Cal selections determined by the system configuration. (The system defaults to all four kV stations, but you can choose kV stations to calibrate during reconfig.)**

- 3.) Run the selected air calibrations.
- 4.) When the calibration process completes, click on QUIT.

Section 5.0

Auto CT N Number Adjustment

Note: If it has been more than 30 minutes since calibration scans, rerun FASTCAL (approx 15 min). N Number Adjustment needs the tube to be warm, or CT numbers will change.

Run the small and large phantom CT N number adjustment before you acquire the image series, or any time you doubt the N# values. The calibration procedure modifies the pcal (phantom cal) vector to assign water a CT number equal to zero. After a successful N number adjustment, any water phantom scanned against a small or large CAL has a CT number near zero.

- 1.) Place the QA Phantom in the holder.
- 2.) Center the Phantom (refer to procedure [2.1.2 on page 48](#))
- 3.) From the application desktop, select SCANNER UTILITIES.
- 4.) Select AUTO CT NUMBER ADJUSTMENT.
- 5.) Follow directions on the screen.
 - a.) When centering the phantom: (refer to
 - * Centering may be easier to perform from the gantry right side.
 - * Center on the water portion of the phantom (not over the internal block).
 - b.) The program will automatically make adjustments. If the program is unable to make adjustments, it will stop at the technique it is unable to adjust. Troubleshoot with manual CT # Adjust before continuing.

Initial: _____

Section 6.0

Tomographic Plane Indication

- 1.) Place the QA phantom on the phantom holder.
- 2.) Center the Phantom (refer to procedure [2.1.2 on page 48](#))
- 3.) Turn ON the internal alignment lights, and drive the phantom into the gantry opening, until the line on the phantom lines up with the internal laser lights.
- 4.) Verify that BOTH internal axial lasers line up along the line on the QA phantom. If not, check table/gantry, cradle, and/or laser alignment.
- 5.) Center the phantom in the scan plane with the calibration program. (See [2.1.2 on page 48](#), for details on the phantom centering procedure.)
- 6.) Select the service protocol, TOMO PLANE INDICATION. (See [2.1.1 on page 48](#), for details on scanning with service protocols.)

or

Manually select the scan parameters in [Table 3-2](#).

Scan Type	kV	mA	SFOV	Thickness	Scan Time	Start Loc.	End Loc.	Algorithm	Interval
Helical	120	200	Small	1.25HQ	1.0sec	I3.0	S3.0	Bone detail	0.2

Table 3-2 Tomographic Plane Indication Scan Parameters

- 7.) Display the image series, and locate the scan plane indicator, the longest bar in the bar pattern on the right side of the phantom. The right side of the phantom corresponds to the side of the image labeled **L** on the display screen.
- 8.) On the HHS Data Sheet, record the scan location (shown on the image annotation) of the image with the darkest scan plane indicator (darkest long bar).
- 9.) If your system meets all the installation and alignment specifications, the image at scan location zero (S0.0) should contain the scan plane indicator. If scan location S1.0 or scan location I1.0 has the darkest bar, the system still meets the specification. The scan plane deviation should equal $S0.0 \pm 1.0\text{mm}$. If necessary, adjust the internal alignment light position to meet the $S0.0 \pm 1.0\text{mm}$ requirement.
- 10.) Repeat the Tomographic Plane Indication test with the external alignment lights.
 - a.) Use the external alignment light, and press the external landmark.
 - b.) Verify the external light lines up along the black line on BOTH the left and right sides of the QA phantom.
 - c.) The scan plane indication must fall within the $S0.0 \pm 1.0\text{mm}$ specification.
- 11.) Initial below.

Initial: _____

Section 7.0 Image Series

IMPORTANT: Run ALL Image Series Tests in **Auto Mode**. Manual procedures are provided in this section as **REFERENCE ONLY**.

7.1 Scan Protocol

The person who acquires the image series has the responsibility to review the images and verify they meet the specifications listed on data sheets. Responsibilities also include means and standard deviation measurements and keeping a record of failures that occur during the image series. Unless otherwise stated, use the following scan parameters during the image series acquisition:

- Scan FOV equal to display FOV (Field of View)
- 512x512 matrix size

Note: Consider **any** image series scan that does not meet specifications as failing.

For means and standard deviations, 90% of the slices must pass. Any failure on a particular technique requires at least ten additional slices to evaluate effectively.

Systems with metal-free cradles have a phantom holder with a perpendicular adjustment (Z-axis) knob on it. **Each time you change phantoms**, make sure you use a bubble level, and the Z-axis knob on the phantom holder, to level the phantom.

7.2 Data Recording: Means and Standard Deviation

Any failure on a particular technique requires at least a ten additional slices to evaluate effectively. For means and standard deviations, 90% of the slices must pass.

- Record means to two decimal places, and round to the nearest one-tenth, (one decimal place) when you compare the resulting values to the spec.
- Record standard deviations to two decimal places, then round off to one decimal place, to compare it to the spec.
- Average standard deviations: Use two decimal places to average the values, then round off to one place.

Before you record the means and standard deviations, check the image data sheets to determine whether to average the means and standard deviations, or record them slice by slice. Make sure you record all the required image data on the HHS data sheets.

7.3 Term Definitions and Screens

Xc - Mean CT number for the specified center coordinates of the phantom image.

AvXc - Average Mean CT number for the center of the phantom image: Average the mean CT value for all specified center coordinates of all slices in an exam.

Xo - Mean CT number for the outside of the phantom image: Average the mean CT value for all specified outside coordinates of one slice.

AvXo - Average outside mean CT number for the number of slices in an exam.

AvSDc - Average image noise about the center image coordinate (measured as the standard deviation) of all slices in an exam.

AvSDo - Average image noise (standard deviation) for the outside of a phantom: Average of all outside coordinates of all the slices in an exam.

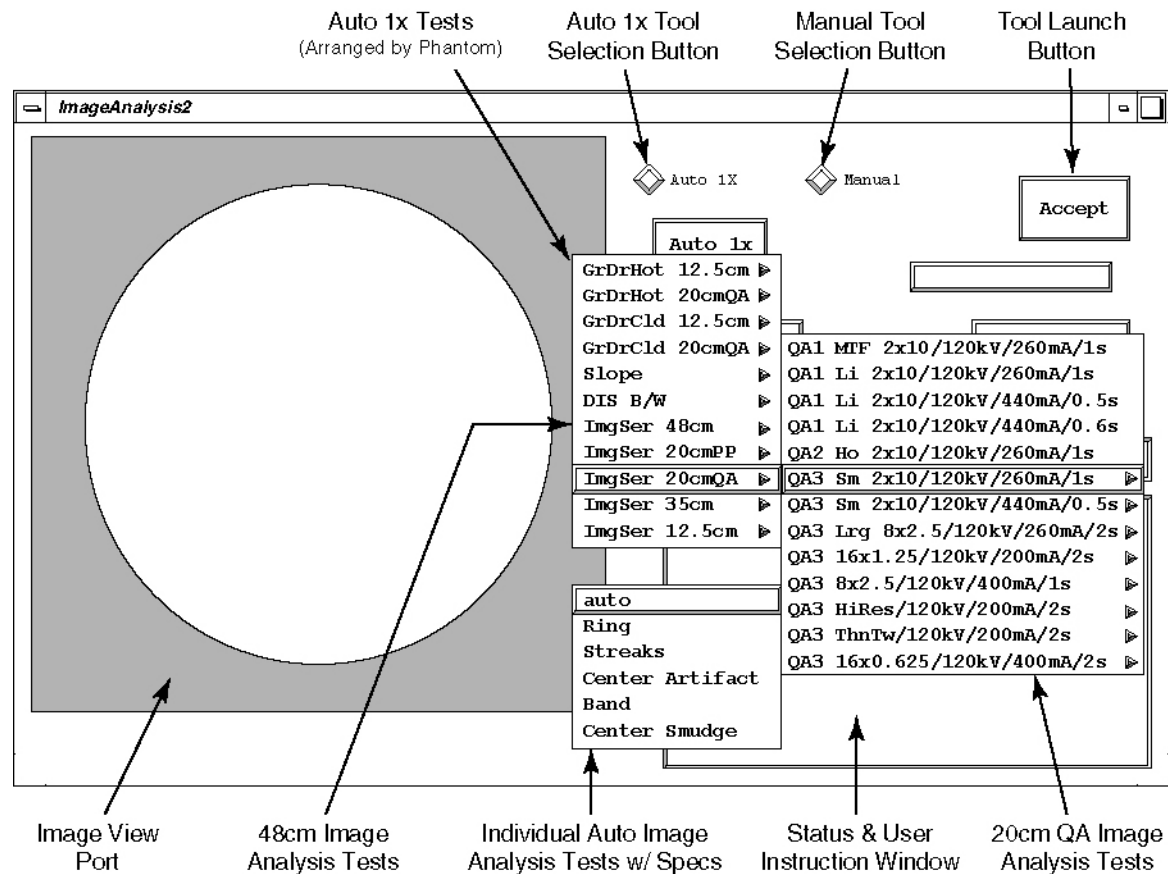


Figure 3-2 Image Analysis Tool User Interface - Auto 1x Test Pull Down Menu

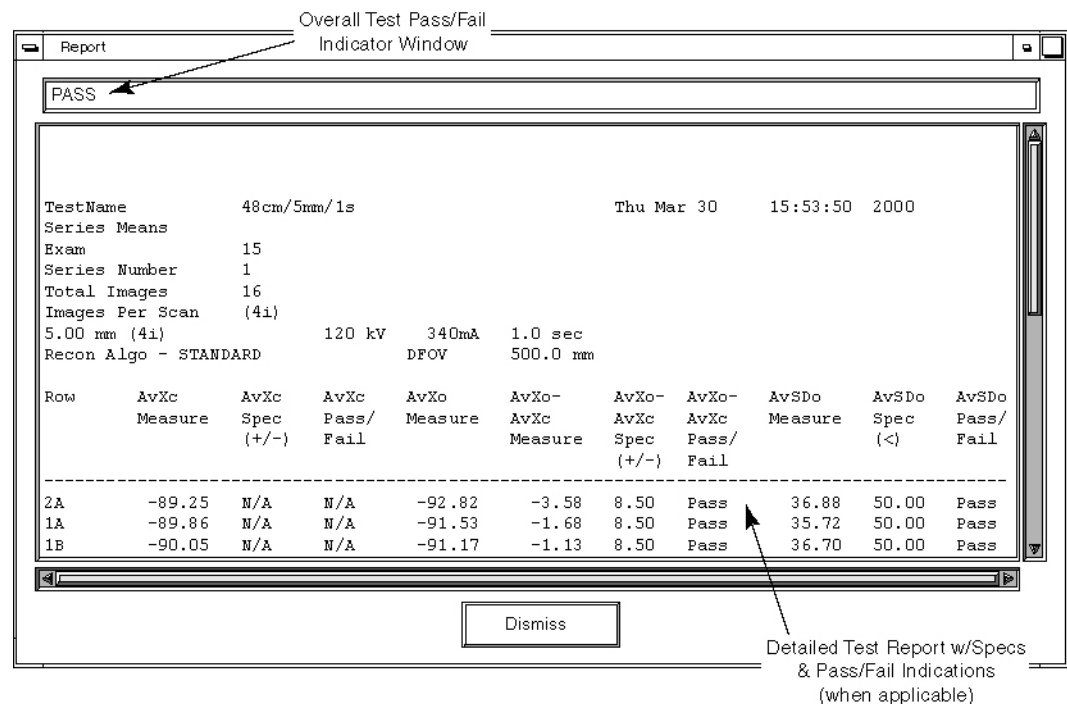


Figure 3-3 Image Analysis Tool User Interface - Test Results Report Window

7.4 48cm Phantom Image Series Image Performance Verification

7.4.1 Acquiring the 48cm Phantom Image Series

- 1.) Mount the Phantom Holder on the head-end of the table.
- 2.) Mount the 48cm Phantom on the Phantom Holder.
- 3.) Align, level, & center the 48cm Phantom.
 - Align phantom using the internal laser lights.
 - Level phantom using bubble level and the Z Axis knob on the Phantom Holder.
 - Center phantom using the CENTER PHANTOM procedure in the left head SCANNER UTILITIES selection and the X and Y Axis knobs on the Phantom Holder.
- 4.) Set up the system to scan a single, 64 image, 48cm Phantom series.

MANUAL SCAN PROTOCOL SETUP

Refer to [Table 3-3](#) and set-up an Axial scan with the parameters shown.

Series Description	Scan Type	Start Loc.	End Loc.	Total # of Images	Thick Speed	Interval (mm)	Gantry Tilt	SFOV	kV	mA	Total Exposure Time	DFOV (cm)	Recon Type
48 16x1.25/120kV /400mA/2s	Axial Full 2.0 sec	19.375	59.375	64	1.2516i	0.00	S0.0	Large	120	400	8.0 sec.	50.0	Std

Table 3-3 48cm Phantom Image Series Scan Parameters

AUTO SCAN PROTOCOL SETUP

- a.) On the Exam Rx desktop, select NEW PATIENT.
 - b.) Type the following entries in the two listed Patient Information fields:
 - * Patient ID: **Service**
 - * Name: **48cm Phantom Image Series**
 - c.) From the Protocol Selection display, click on the Service Tab under Anatomical Selector.
 - d.) On the Service Protocol Selection window, click on MANUFACTURING.
 - e.) On the Service Category Protocol List, click on 45.7 IMGSER 48CM MDAS 16 16.
 - f.) On the ExamRx protocol parameter display, skip to the 3rd Series (Series Description: 48 16x1.25/120kV/400mA/2s).
 - g.) Set internal Landmark.
- 5.) Acquire a single, 64 image, 4 scan, image series of the 48 cm Phantom.

7.4.2 Image Performance Verification

7.4.2.1 Series Means (AvXo-AvXc and AvSDo) & Center Smudge

- 1.) Launch the Image Analysis Tool.
 - a.) From the Global Control Palette, click on the Service Desktop.
 - b.) From the Service Desktop upper navigational bar, click on the IMAGE QUALITY icon.
 - c.) In the left navigational area directory, click on IMAGE ANALYSIS.

Note: During the Image Analysis process, you will need to switch between the Service Browser screen (to select the image series to analyze) and the Image Analysis screen (to run a particular Auto 1x analysis test). Use ALT/F1 to bring a window to the front. Use ALT/F3 to send a window to the back.

- 2.) Using the AUTO 1X Tool on the Image Analysis screen, generate a Series Means and Center Smudge report for the 1st series of the 48cm Phantom acquisition.
 - a.) From the Service Browser window, select the 48cm Phantom series acquired in Section 7.4.1.
 - b.) On the Image Analysis Tool window, click on the AUTO 1X diamond shaped button. (See [Figure 3-2, on page 54.](#))
 - c.) Click on the AUTO 1X rectangular button, select the IMGSER 48CM test from the pull-down menu, slide cursor to the right and select 16X1.25/120KV/400MA/2S from the test pull-down menu, and continue to slide cursor to the right to select AUTO.
 - d.) Click on the ACCEPT button.
- 3.) Review the Image Analysis Report data and record the Series Means (AvXo-AvXc, and AvSDc) data and the Center Smudge data for each row in [Table 3-4](#).
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS. (See [Figure 3-3, on page 54.](#))
 - b.) Verify the scan parameter values displayed in the report match those in [Table 3-3](#).
 - c.) Record the Series Means (AvXo-AvXc, and AvSDc) data and the Center Smudge data for each row in [Table 3-4](#).

7.4.2.2 Band and Streak Artifact

- 1.) Review the 48cm Phantom images acquired in Section 7.4.1 for Bands and Streaks using the Image Works Desktop.
 - a.) From the Global Control Palette, click on the IMAGE WORKS Desktop.
 - b.) From the Image Works Browser window, select the 48cm Phantom series acquired in Section 7.4.1.
 - c.) From the Image Works Browser window, click on the VIEWER button.
 - d.) While paging through the 64 images in the series, analyze images for any Band or Streak artifacts. Adjust Window/Level using the center mouse button, as required.
 - e.) Note any image that appears to have a Band or Streak artifact.
 - 2.) Verify any images noted above (step 1e) meet 48cm Phantom Band or Streak Image Performance requirements.
 - a.) From the Global Control Palette, click on the Service Desktop.
 - b.) From the Service Browser window, select an image of the 48cm Phantom series acquired in Section 7.4.1 that appeared to have a Band or Streak while reviewing the series in Image Works.
 - c.) On the Image Analysis Tool window, click on the AUTO 1X rectangular button, select the IMGSER 48CM test from the pull-down menu, slide cursor to the right and select 16X1.25/120KV/400MA/2S from the test pull-down menu, and continue to slide cursor to the right to select BAND or STREAK.
 - d.) Click on the ACCEPT button.
- Note: If required, adjust Window/Level using the center mouse button.
- Note: Position and size the Band or Streak ROI using the left cursor button. To ensure specifications are properly applied, do not adjust the Reference ROI's for the Band and Streak artifact tests. The Reference ROI's are adjustable after the 1st Accept Modification click.
- f.) Click on ACCEPT MODIFICATION twice, to generate a report.
 - g.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS.
 - h.) Repeat steps b through g, for each image noted in Step 1e.

Box Size = 1600 mm² to 2500 mm²
 45 mm (+/- 5 mm) x 45 mm (+/- 5 mm)
 45 (+/- 4 pixels) x 45 (+/- 4 pixels)

Box Positions: Box 1 = 0 mm x 0 mm
 Box 2 = 0 mm x -190 mm
 Box 3 = 190 mm x 0 mm
 Box 4 = 0 mm x 190 mm
 Box 5 = -190 mm x 0 mm

Row	Images	Brightness Uniformity (AvXo - AvXc)	Noise (AvSDo)	Center Smudge Row Values	Comments
8A	1, 17, 33, 49				
7A	2, 18, 34, 50				
6A	3, 19, 35, 51				
5A	4, 20, 36, 52				
4A	5, 21, 37, 53				
3A	6, 22, 38, 54				
2A	7, 23, 39, 55				
1A	8, 24, 40, 56				
1B	9, 25, 41, 57				
2B	10, 26, 42, 58				
3B	11, 27, 43, 59				
4B	12, 28, 44, 60				
5B	13, 29, 45, 61				
6B	14, 30, 46, 62				
7B	15, 31, 47, 63				
8B	16, 32, 48, 64				
Specifications		< +/- 17.0	< 50	< +/- 14	

Table 3-4 48cm Phantom CT# Brightness Uniformity, Noise & Center Smudge Row Performance Data Sheet

7.4.3 Failure Recommended Actions

7.4.3.1 Series Means (AvXo-AvXc or AvSDo) Failure Recovery

Specifications

Each Row of the series must pass 48cm Series Means (for the first series scan parameters) specifications:

- AvXo - AvXc: < +/- 17.0
- AvSDo: < 50.0

Recommended Recovery

- 1.) Repeat Detailed Cal for the 48cm Phantom.
- 2.) Repeat Sections 7.4.1 & 7.4.2, to verify Image Performance.

7.4.3.2 Center Smudge Failure Recovery

Specifications

80% the Rows must pass Center Smudge specifications when re-scanning:

Center Smudge: Smudge Factor < +/- 14.0

Recommended Recovery

- 1.) If one image of one Row fails (Row fails once):
 - Re-scan the 48cm Phantom acquiring two series of 64 images using the same protocol settings as listed in [Table 3-3](#) (on [page 55](#)).
 - Analyze the two 64 re-scanned image series using the Image Analysis IMGSER 48CM -> 16X1.25/120KV/400MA -> AUTO tool.
 - Verify that all the 128 re-scanned images do not fail Series Means (AvX_O - AvX_C and AvS_D) or Center Smudge.
- 2.) If more than one image of one Row fails (Row fails more than once) or multiple rows are failing:
 - Repeat Detailed Cal for the 48cm Phantom.
 - Re-scan the 48cm Phantom acquiring three series of 64 images using the same protocol settings as listed in [Table 3-3](#) (on [page 55](#)).
 - Analyze the three re-scanned image series using the Image Analysis IMGSER 48CM -> 16X1.25/120KV/400MA -> AUTO tool.
 - Verify that all the 192 re-scanned images do not fail Series Means (AvX_O - AvX_C and AvS_D) or Center Smudge.

7.4.3.3 Band or Streak Artifact Failure Recovery

Specifications

- Band Artifact: Band Factor < +/- 8.0
 - Streak Artifact: Streak Factor < +/- 4.0
-

Recommended Recovery

- 1.) Troubleshoot Band or Streak artifact failures, as described in the *System Service Manual*.
- 2.) Repeat Detailed Cal for the 48cm Phantom.
- 3.) Repeat Sections 7.4.1 & 7.4.2, to verify Image Performance.

7.5 20cm QA Phantom Image Series Image Performance Verification

7.5.1 Acquiring the 20cm QA Phantom Image Series

- 1.) Mount the Phantom Holder on the head-end of the table.
- 2.) Mount the 20cm QA Phantom on the Phantom Holder.
- 3.) Align, level, & center the 20cm QA Phantom.
 - Align black line on phantom using the internal laser lights.
 - Level phantom using bubble level and the Z Axis knob on the Phantom Holder.

- Center phantom using the CENTER PHANTOM procedure in the left head SCANNER UTILITIES selection and the X and Y Axis knobs on the Phantom Holder.
- 4.) Set up the system to scan three 20cm QA Phantom image series with a Recon of the 1st series.

MANUAL SCAN PROTOCOL SETUP

Refer to [Table 3-5](#) and set-up an Axial scan with the parameters shown.

Series Description	Scan Type	Start Loc.	End Loc.	Total # of Images	Thick Speed	Interval (mm)	Gantry Tilt	SFOV	kV	mA	Total Exposure Time	DFOV (cm)	Recon Type
QA1 MTF 2x10/120kV/ 260mA/1s	Axial Full 1.0 sec.	I5.00	S5.00	4	10.02i	0.00	S0.0	Small	120	260	2.0 sec.	25.0	Std
Recon 2: Q	Axial Full 1.0 sec.	I5.00	S5.00	4	10.02i	0.00	S0.0	Small	120	260	2.0 sec.	15.0	Bone
QA2 Ho 2x10/120kV/ 260mA/1s	Axial Full 1.0 sec.	S40.00	S50.00	8	10.02i	0.00	S0.0	Small	120	260	4.0 sec.	25.0	Std
QA3 Sm 2x10/120kV/ 260mA/1s	Axial Full 1.0 sec.	S55.00	S65.00	8	10.02i	0.00	S0.0	Small	120	260	4.0 sec.	25.0	Std

Table 3-5 20cm QA Phantom Image Series Scan Parameters

AUTO SCAN PROTOCOL SETUP

- a.) On the Exam Rx desktop, select NEW PATIENT.
- b.) Type the following entries in the listed Patient Information following fields:
 - * Patient ID: **Service**
 - * Name: **20cm QA Phantom Image Series**
- c.) From the Protocol Selection display, click on the Service Tab under Anatomical Selector.
- d.) On the Service Protocol Selection window, click on MANUFACTURING.
- e.) On the Service Category Protocol List, click on the 45.9 IMGSER 20CMQA MDAS_16_16 selection.
- f.) On the ExamRx protocol parameter display, select the 1st Series (Series Description: QA1 MTF 2x10/120kV/260mA/1s).
- g.) Set internal Landmark.
- 5.) Acquire the 1st 20cm QA Phantom image series and one Recon series (MTF and Contrast, Visible Lines).
- 6.) Acquire the 3rd 20cm QA Phantom image series (Series Description: QA2 Ho 2x10/120kV/260mA/1s).
- 7.) Acquire the 4th 20cm QA Phantom image series (Series Description: QA3 Sm 2x10/120kV/260mA/1s).

7.5.2 20cm QA Phantom Image Series Image Performance Verification

1st QA Phantom Image Series (4 Image MTF Average and Contrast Scale) Image Performance Verification

- 1.) Launch the Image Analysis Tool.
 - a.) From the Global Control Palette, click on the Service Desktop.
 - b.) From the Service Desktop upper navigational bar, click on IMAGE QUALITY TESTS.

- Note:
- c.) In the left navigational area directory, click on IMAGE ANALYSIS.
During the Image Analysis process, you will need to switch between the Service Browser screen (to select the image series to analyze) and the Image Analysis screen (to run a particular Auto 1x analysis test). Use ALT/F1 to bring a window to the front. Use ALT/F3 to send a window to the back.
 - 2.) Using the Auto 1x Tool on the Image Analysis screen, generate a QA#1 MTF and QA#1 Contrast Scale report for the 1st series of the 20cm QA Phantom acquisition.
 - a.) From the Service Browser window, select the 1st series of the 20cm QA Phantom series acquired in section 7.5.1.
 - b.) On the Image Analysis Tool window, click on the AUTO 1X diamond shaped button. (See [Figure 3-2, on page 54.](#))
 - c.) Click on the AUTO 1X rectangular button, select the IMGSER 20CMQA test from the pull-down menu, slide cursor to the right to select QA1 MTF 2X10/120KV/260MA/1S from the test pull-down menu.
 - d.) Click on the ACCEPT button.
 - 3.) Review the Image Analysis Report data and record the Per Image MTF, 4 Image MTF Average, and Per Image Contrast Scale data for each row in Table 10-7.
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS. (See [Figure 3-3, on page 54.](#))
 - b.) Verify the scan parameter values displayed in the report match those in [Table 3-5 \(on page 59\)](#) for the 1st 20cm QA Image Series.
 - c.) Record Per Image MTF and Per Image Contrast Scale data for each image in [Table 3-6.](#)
 - d.) Record the 4 Image MTF Average in [Table 3-6.](#)

Image	MTF	MTF 4-slice average	Contrast Scale	Comments
1		N.A.		
2		N.A.		
3		N.A.		
4		N.A.		
Specifications	N.A.	0.65 to 1.0	110.0 to 130.0	N.A.

Table 3-6 20cm QA#1 Phantom High Contrast Spatial Resolution Image Performance (MTF and Contrast Scale)

2nd QA Phantom Image Series (4 Bone Retro Image Visible Lines) Image Perf. Verification

- 1.) Using the Auto 1x Tool on the Image Analysis screen, generate a Per Image QA#1 Lines report for the 2nd series of the 20cm QA Phantom acquisition.
 - a.) From the Service Browser window, select the 1st image of the 2nd series of the 20cm QA Phantom series acquired in section 7.5.1 (on [page 58](#)).
 - b.) Click on the AUTO 1X rectangular button, select the IMGSER 20CMQA test from the pull-down menu, and slide cursor to the right to select QA1 LI 2X10/120KV/260MA/1S.
 - c.) Click on the ACCEPT button. The tool displays a Visible Lines pop-up window.
 - d.) On the Visible Lines pop-up window, click on the VISIBLE LINE button.
- Note:
- The Image Analysis Tool automatically sets the Window/Level for optimal viewing. If required, adjust Window/Level using the center mouse button.
 - e.) From the Visible Line pull-down menu, select the letter (A through F) that matches the smallest line pair pattern that you can discern in the Image View Port.
 - f.) On the Visible Lines pop-up window, click on the OK button.

- 2.) Review the Image Analysis Report data and record the Per Image QA#1 Visible Lines data for the 1st image in [Table 3-7](#).
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS. (Refer to [Figure 3-3, on page 54.](#))
 - b.) Verify the scan parameter values displayed in the report match those in [Table 3-5](#) (on [page 59](#)) for the 2nd 20cm QA Image Series.
 - c.) Record the Per Image QA#1 Visible Lines data for the 1st image in [Table 3-7](#).
- 3.) Repeat Steps 1 and 2, for each of the remaining three images in the 2nd 20cm QA Image Series.

Image	Line Patterns Visible	Comments
1		
2		
3		
4		
Specifications	B, C, D, E, F	N.A.

Table 3-7 20cm QA#1 Phantom High Contrast Spatial Resolution Image Performance (Visible Lines)

3rd QA Phantom Image Series (4 Image Visible Holes) Image Performance Verification

- 1.) Using the Auto 1x Tool on the Image Analysis screen, generate a Per Image QA#2 Holes report for the 3rd series of the 20cm QA Phantom acquisition.
 - a.) From the Service Browser window, select the 1st image of the 3rd series of the 20cm QA Phantom series acquired in section 7.5.1.
 - b.) Click on the AUTO 1X rectangular button, select IMGSER 20CMQA from the pull-down menu, and slide cursor to the right to select QA2 HO 2X10/120KV/260MA/1S.
 - c.) Click on the ACCEPT button. The tool displays a Visible Hole pop-up window.
 - d.) On the Visible Hole pop-up window, click on the VISIBLE HOLE button.
 - e.) From the Visible Hole pull-down menu, select the number (1 through 5) that matches the smallest hole that you can discern in the pattern centered in the image displayed in the Image View Port.
 - f.) On the Visible Hole pop-up window, click on the OK button.
- 2.) Review the Image Analysis Report data and record the Per Image QA#2 Visible Holes data for the 1st image in [Table 3-8](#).
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS. (Refer to [Figure 3-3, on page 54.](#))
 - b.) Verify the scan parameter values displayed in the report match those in [Table 3-5](#) for the 3rd 20cm QA Image Series.
 - c.) Record the Per Image QA#2 Visible Holes data for the 1st image in [Table 3-8](#).

Note:

The Image Analysis Tool automatically sets the Window/Level for optimal viewing. If required, adjust Window/Level using the center mouse button.

- 3.) Repeat Steps 1 and 2 for the 3rd, 5th, and 7th images in the 3rd 20cm QA Image Series.

Image	Visible Holes Viewable at Window 20	Contrast Factor	Comments
1			
3			
5			
7			
Specifications	See Table 6-9	2.0 to 12.0	N.A.

Table 3-8 20cm QA#2 Phantom Low Contrast Detectability Image Performance (Visible Holes)

Contrast Factor Range (Box 1 Means - Box 2 Means)	Visible Number of Holes		Smallest Visible Hole Size
	Lower Limit *	Upper Limit *	
2.00 to 3.99	2	5	7.5 mm
4.00 to 7.99	3	5	5.0 mm
8.00 to 12.00	4	5	3.0 mm
* Required Number of Visible Holes depends on the Contrast Factor			

Table 3-9 20cm QA#2 Phantom Visible Hole Specifications

4th QA Phantom Image Series (QA#3 Small) Image Performance Verification

- 1.) Using the Auto 1x Tool on the Image Analysis screen, generate a QA#3 Small report for the 4th series of the 20cm QA Phantom acquisition.
 - a.) From the Service Browser window, select the 4th series of the 20cm QA Phantom series acquired in section 7.5.1 (on [page 58](#)).
 - b.) Click on the AUTO 1X rectangular button, select IMGSER 20CMQA from the pull-down menu, slide cursor to the right to select QA3 SM 2X10/120KV/260MA/1S from the test pull-down menu, and continue to slide cursor to the right to select AUTO. (Refer to [Figure 3-3, on page 54.](#))
 - c.) Click on the ACCEPT button.
- 2.) Review the Image Analysis Report data and record the QA3 Small data for the 4th 20cm QA Image Series in [Table 3-10](#).
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS. (Refer to [Figure 3-3, on page 54.](#))
 - b.) Verify that the scan parameter values displayed in the report match those in [Table 3-5](#) (on [page 59](#)) for the 4th 20cm QA Image Series.
 - c.) Record the QA3 Small data (Row 2A1A and Row 2B1B Series Means $\bar{A}_{VXC}$ and $\bar{A}_{VXO}$ - $\bar{A}_{VXC}$), QA3 ($\bar{A}_{VSDC}$), and the Center Smudge Row data in [Table 3-10](#).

Box Size = 196 mm² to 256 mm²
 15 mm (+/- 1 mm) x 15 mm (+/- 1 mm)
 31 (+/- 2 pixels) x 31 (+/- 2 pixels)

Box Positions: Box 1 = 0 mm x 0 mm
 Box 2 = 0 mm x -80 mm
 Box 3 = 80 mm x 0 mm
 Box 4 = 0 mm x 80 mm
 Box 5 = -80 mm x 0 mm

Row	Images	AvXc	AvXo	AvXo - AvXc	AvSDo	AvSDc	Center Smudge Row Value	Comments
2A1A	1, 3, 5, 7				N.A.			
1B2B	2, 4, 6, 8				N.A.			
Specifications		+3.0 to -3.0	N.A.	+3.0 to -3.0	N.A.	+ 2.9 to +3.5	+2.2 to -2.2	

Table 3-10 20cm QA#3 Phantom CT#, Brightness Uniformity, Center Noise & Center Smudge Row Performance Data Sheet

- 3.) Review the 4th 20cm QA Phantom images acquired in Section 7.5.1 for Rings, Bands, and Streaks using the Image Works Desktop.
 - a.) From the Global Control Palette, click on the IMAGE WORKS DESKTOP.
 - b.) From the Image Works Browser window, select the 4th 20cm QA Phantom series acquired in section 7.5.1.
 - c.) From the Image Works Browser window, click on the VIEWER button.
 - d.) While paging through the eight images in the series, analyze images for any Ring, Band, or Streak artifacts. Adjust Window/Level using the center mouse button, as required.
 - e.) Note any image that appears to have a Ring, Band, or Streak artifact.
 - 4.) Verify any images noted above (step 3e) meet 20cm QA Phantom Ring, Band, or Streak Image Performance requirements.
 - a.) From the Global Control Palette, click on the Service Desktop.
 - b.) From the Service Browser window, select an image of the 4th 20cm QA Phantom series acquired in section 7.5.1 that appeared to have a Ring, Band, or Streak while reviewing the series in Image Works.
 - c.) On the Image Analysis Tool window, click on the AUTO 1X rectangular button, select the IMGSER 20CMQA test from the pull-down menu, slide cursor to the right to select QA3 SM 2X10/120KV/260MA/1S from the test pull-down menu, and continue to slide cursor to the right to select RING, BAND, or STREAK.
 - d.) Click on the ACCEPT button.
- Note: If required, adjust Window/Level using the center mouse button.
- e.) Position and size the Ring, Band, or Streak ROI, using the left cursor button.
- Note: To ensure specifications are properly applied, do not adjust the Reference ROIs for the Ring, Band, and Streak artifact tests. The Reference ROIs are adjustable after the 1st Accept Modification click.
- f.) Click on ACCEPT MODIFICATION twice to generate a report.
 - g.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS.
 - h.) Repeat steps b through g for each image noted in Step 3e.

7.5.3 Failure Recommended Actions

7.5.3.1 1st Image Series (4 Image MTF Average) Failure Recovery

Specifications

The 4 Image MTF Average must pass specifications:

4 Image MTF Average: 0.65 to 1.00

Recommended Recovery

- 1.) Repeat Detailed Cal for the 20cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 20cm QA Phantom.
- 3.) Repeat Sections 7.5.1 & 7.5.2, to verify Image Performance.

7.5.3.2 2nd Image Series (Visible Lines) Failure Recovery

Specifications

The Largest 5 line pair patterns (coded F, E, D, C, and B) must be visible for each of the four images in this series.

Recommended Recovery

Note: The most common failure for this test is that the phantom has air bubbles that are obscuring the line pair patterns.

- 1.) Carefully inspect the 20cm QA Phantom for air bubbles. If required, refill the phantom with water, to eliminate all air bubbles.
- 2.) Repeat Sections 7.5.1 & 7.5.2, to verify Image Performance.

7.5.3.3 3rd Image Series (Visible Holes) Failure Recovery

Specifications

The number of visible holes in this image series must pass specifications as follows:

- Contrast Factor = 2.0 to 3.99:
Minimum of 2 holes must be visible (10mm & 7.5mm)
 - Contrast Factor = 4.0 to 7.99:
Minimum of 3 holes must be visible (10mm, 7.5mm & 5mm)
 - Contrast Factor = 8.0 to 12.0:
Minimum of 4 holes must be visible (10mm, 7.5mm, 5mm, & 3mm)
-

Recommended Recovery

- 1.) Repeat Detailed Cal for the 20cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 20cm QA Phantom.

- 3.) Repeat Sections 7.5.1 & 7.5.2, to verify Image Performance.

7.5.3.4 4th Image Series Failure Recovery

Series Means

SPECIFICATIONS

Both Rows (2A1A, 1B2B) of the series must pass QA#3 Series Means specifications:

- AvXC : $< \pm 3.0$
- $\text{AvXO} - \text{AvXC}$: $< \pm 3.0$

RECOMMENDED RECOVERY

- 1.) Repeat Detailed Cal for the 20cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 20cm QA Phantom.
- 3.) Repeat Sections 7.5.1 & 7.5.2, to verify Image Performance.

QA#3 (AvSDc)

SPECIFICATIONS

Both Rows (2A1A, 1B2B) of the series must pass QA#3 AvSDc specifications:

- AvSDc (< 5K Scans): 2.90 to 3.50
- AvSDc (> 5K Scans): 2.80 to 3.60

RECOMMENDED RECOVERY

- 1.) Repeat Detailed Cal for the 20cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 20cm QA Phantom.
- 3.) Repeat Sections 7.5.1 & 7.5.2, to verify Image Performance.

Center Artifact or Center Smudge

SPECIFICATIONS

All eight images of the series must pass 20cm QA#3 Center Artifact and Center Smudge specifications:

- Center Artifact Factor: ± 3.5
- Center Smudge Factor: ± 2.2

RECOMMENDED RECOVERY

- 1.) Repeat Detailed Cal for the 20cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 20cm QA Phantom.
- 3.) Repeat Sections 7.5.1 & 7.5.2, to verify Image Performance.

Ring, Band, or Streak Artifact Failure Recovery

SPECIFICATIONS

- Ring Artifact: Ring Factor < +/- 4.8
- Band Artifact: Band Factor +/- 2.8 from 0cm to 8.5cm
< +/- 2.8 > 8.5cm
- Streak Artifact: Streak Factor < +/- 4.0

RECOMMENDED RECOVERY

- 1.) Troubleshoot Ring, Band or Streak artifact failures as described in the Service Manual.
- 2.) Repeat Detailed Cal for the 20cm QA Phantom.
- 3.) Repeat Sections 7.5.1 & 7.5.2, to verify Image Performance.

Section 8.0

System Functional Test

Use the system tests in the following sections to exercise all aspects of the system and to ensure system integrity before releasing to the customer. Although the means, standard deviation, and resolution specifications do not apply during system functional tests, treat any artifact or image anomaly as a failure.

If you encounter a failure during the system tests:

- Record any evidence of artifacts, such as rings, streaks, shading, cupping, noise, or center artifacts.
- Correct artifacts, system test, or image series failures when they occur. Any delay in repairs could increase the number of retests.

- 1.) Place the QA phantom on the cradle.
 - Drive the table to an elevation of 100.
 - Align the line on the phantom with the internal laser lights.

Never scan above 50mA without first placing a phantom in the field of view. Levels in excess of 50mA can cause temporary radiation damage to the detector that lasts several hours. If you acquire image series cals with a radiation damaged detector, the cals may cause artifacts in subsequent image series scans.

- 2.) Select the service protocol, SYSTEM SCAN/CUST QUAL RELI.

Note:

Stop the service protocol after the second helical series. It is not necessary to proceed past that point in the protocol.

-or-

Manually select the scan parameters specified in [Table 3-11](#).

Scan Type	kV	mA	SFOV	Thickness	Rotation Time	Start Loc.	End Loc.	Tilt/Table Speed
Scout	120	40	-	-	-	S200	I800	0°
Scout	120	40	-	-	-	S200	I800	90°
Axial (2 scans)	120	50	Large	10mm/1i	1.0sec	S0	S0	I30
Axial (2 scans)	120	50	Large	10mm/1i	1.0sec	S0	S0	S30
Helical HQ	120	50	Large	5mm	1.0sec	S70	I75	15mm/rot
Helical HS	120	50	Large	5mm	0.8sec	S70	I75	22.5mm/rot
Air	120	50		16 x 1.25	1.0sec			
Air	120	50		8 x 2.5	1.0sec			

Table 3-11 System Functional Test Scan Parameters

- 3.) Review all air scans for presence of any artifacts (rings, bands, streaks, etc.).
- 4.) If any artifacts are present, review the MSD Plots for each AIR scan, using Diagnostic Data Analysis.
- 5.) Complete the scans.

Initial: _____

Section 9.0

Save System State

Use the following commands to create the System State disk (DVD or MOD).

- 1.) Load the archive medium (DVD or MOD) into the archive drive, in the SCSI tower.

- 2.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon, .

- 3.) Click on the PM icon.



- 4.) Select SYSTEM STATE to open the System State Save/Restore menu.

- 5.) Select ALL

- 6.) Select SAVE

- 7.) When the save operation completes, select FILE and QUIT from the pull down menu.

- 8.) Remove the archive medium (DVD or MOD) from the drive.

Chapter 4

Customer Options Installation & Verification

Note: Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

CT Options

- ☐ Injector functional tests completed.
- ☐ AWW functional tests completed.
- ☐ Filming/Camera/DASM functional tests completed.
- ☐ UPS functional tests completed.
- ☐ Network items installed and functional tests completed.
- ☐ Verify that all customer software options are installed and functional.
 - * Direct 3D
 - * SmartScore Pro
 - * CT Perfusion
- ☐ Teleradiology Connections (Refer to [Section 10.0 on page 98](#))
- ☐ Auxiliary Monitor
 - * Flat Panel
 - * CRT

Section 2.0

Install Options

Refer to:

- GE Prints and schematics for mechanical (physical) location of option
- FDO shipment for identification of items
- Installation Specialist for installation instructions if they differ from print

2.1 Camera (Filming Device)

Refer to documentation shipped with camera

- Chapter 3 of this manual
- DICOM 2210573 GE Document
- DICOM Print 2152913

2.2 Injector

Refer to documentation shipped with the injector.

Complete a Functional test of the injector.

- The position of the ceiling mount assembly must allow the ceiling stand movement and gantry to tilt, typically 20 degrees, without interference.

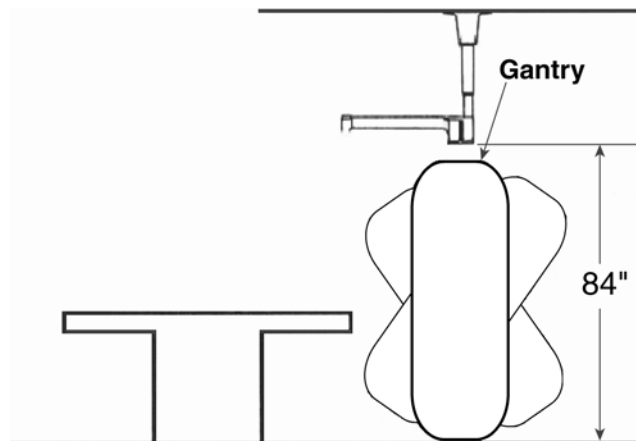


Figure 4-1 Ceiling Mount Position

2.3 Advantage Windows Workstation (AWW)

Refer to the directions provided:

- Pre-install 2111833
- Service 2111831

2.4 Uninterruptible Power System (UPS)

Refer to directions shipped with uninterruptible power system (UPS):

- Powerware B7999PS
- Liebert B7999PT
- Installation support phone numbers found in shipped documentation or in *LightSpeed 4.X Pre-Installation Manual*.

2.5 Other Options

Refer to CT-2211224-100 *LightSpeed Options* for all other options.

Section 3.0

DICOM Network Introduction

LightSpeed systems support two basic Networking Protocols:

- Advantage NET (IC, Signa 4.X, CT-HLA, CT/I ...)
- DICOM (CT/I, CT Synergy, Advantage Workstations, ...)

DICOM NETWORK INTRODUCTION

DICOM networks basically operate on the **tasks** or services that various devices on the network use or provide. These services are labeled as Application Entity Titles (AE Titles). The LightSpeed 4.X system is a **user** of six DICOM Network Services and is a **provider** of two DICOM Services:

AS A DICOM SERVICE USER:

- Send or **Push** images to another network device.
- Send or **Push** images to a DICOM Printer.
- Review image database on another device and retrieve or **Pull** selected images from that device (Query/Retrieve User).
- Send or **Push** images to a an image storage device and obtain confirmation that the images have been archived (Storage Commitment).
- Obtain Patient Worklist Information from the Hospital HIS/RIS System.
- Store images on MOD media.

AS A DICOM SERVICE PROVIDER:

- Receive **Pushed** images from another network device
- Allow another network device to review the CT image database and to retrieve or **Pull** selected images (Query/Retrieve Provider)

For each DICOM Service that the CT system will be a **User** (except for storing images on MOD media), you must **declare** this device on the CT system using three menu selections. For some devices, you must declare not only the device, but each service (AE Title) that the device provides. For example, you may be required to declare a PACS System twice on the CT system: once as a destination to **push** images and, second, as destination that provides storage commitment capability after images have been **pushed**.

For each DICOM Service that the CT system will be a **Provider**, you must **declare** the CT system on the network device that will be using these services.

Information required to complete configuring a hospital DICOM network is provided by the hospital network administrator (hostnames, IP Addresses) and the DICOM Conformance Statement document (AE Titles, Port Numbers) provided with each DICOM compatible network device on the network.

Section 4.0 Before You Start

4.1 Network Physical Requirements

Before setting up the CT system on the hospital network, verify the following physical items are complete:

- CT console, monitor, keyboard, and mouse are installed and connected.
- CT system power is on
- Hospital Ethernet network RJ45 Class IV twisted pair cable is connected to the CT console network receptacle.
- Hospital network connection is operational and is running 10baseT or 100baseT.

4.2 Network Identity Information

To declare the CT system on the network, ensure the following network identity information is available from the Hospital Network Administrator:

- Hostname (No more than 16 Characters)
- Internet Protocol (IP) Address
- Subnet Net Mask IP Address (if applicable)
- Broadcast Address (if applicable)

4.3 LightSpeed to DICOM Remote Hosts Network Information

To declare DICOM remote hosts (PACS systems, archival devices, review workstations) on the LightSpeed 4.X system, ensure the following information is available for each remote host:

From the Hospital Network Administrator:

- Hostname
- Internet Protocol (IP) Address
- Network Protocol (DICOM for LightSpeed Systems)

From the Remote Host Device DICOM Conformance Statement Document:

- DICOM Application Entity Title or AE Title (DICOM service that remote host provides or uses)
- DICOM Listening Port Number

4.4 LightSpeed to DICOM HIS/RIS Interface Network Information

To declare DICOM Hospital HIS/RIS Interface devices (Mitra and others) on the LightSpeed 4.X System, ensure the following information is available:

From the Hospital Network Administrator: *Internet Protocol (IP) Address*

From the HIS/RIS Interface Device DICOM Conformance Statement Document:

- DICOM Application Entity Title or AE Title (DICOM Service that the HIS/RIS interface provides)
- DICOM Listening Port Number

4.5 LightSpeed to DICOM Printers Network Information

To declare DICOM on the LightSpeed 4.X System, ensure the following information is available for each printer:

From the Hospital Network Administrator:

- Hostname
- Internet Protocol (IP) Address

From the Printer DICOM Conformance Statement Document:

- DICOM Application Entity Title or AE Title (DICOM service that remote host provides or uses)
- DICOM Listening Port Number

Section 5.0

Declaring the System on the Hospital Network

5.1 Enter Configuration Routine

- 1.) On the OC, open a Unix shell window.
- 2.) Enter root as a superuser:
Type: `su -` **ENTER** at the prompt.
Type: `#bigguy` **ENTER** at the password prompt.
- 3.) Change directory to scripts:
Type: `cd /usr/g/scripts` **ENTER** at the root prompt.
- 4.) Launch the Install Utility:
Type: `reconfig` **ENTER** at the prompt.
The OC displays the Install Utility Window as shown in [Figure 4-2](#).

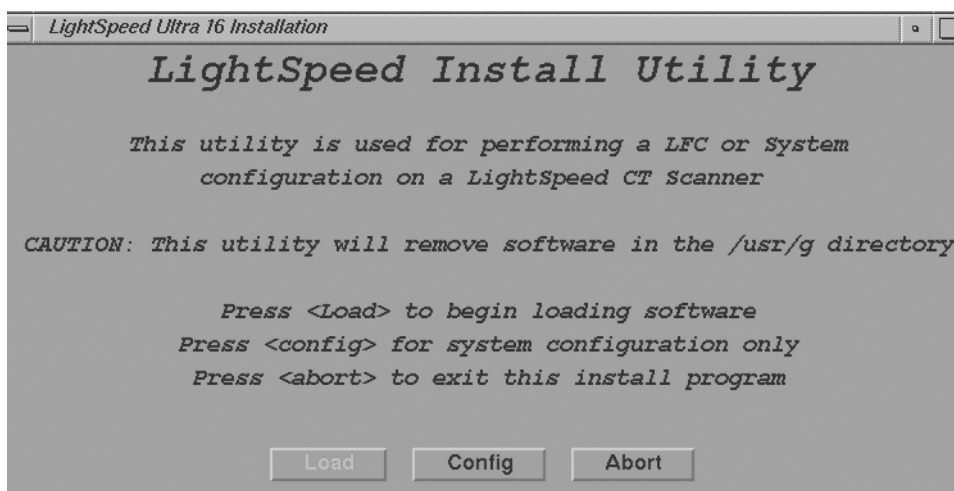


Figure 4-2 Install Utility Window

- 5.) Enter the Configuration Routine:

Using the mouse, click on the CONFIG button.

The OC displays the LightSpeed 4.X System Configuration - System Settings screen, as shown in [Figure 4-3](#).

Figure 4-3 System Settings Screen

5.2 Configure LightSpeed 4.X Network Settings

This screen provides the ability to declare the CT system on a hospital network. Key information such as Host Name, IP Address, Net Mask (for CT systems on a subnet) must be obtained from the hospital network administrator.

- 1.) Select the NETWORK button to display the Network Settings screen as shown in [Figure 4-4](#).
- 2.) Enter the Suite Name.

The Suite Name is a means of identifying this particular CT system as a part of a group of CT Systems in a suite configuration. This Suite Name will appear on all image headers.

The Suite Name must start with a letter, followed by three alphanumeric characters (total MUST be four characters long). The name of the OC interface will be <Suite Name>_oc and the SBC interface will be <Suite Name>_sbc.

The screenshot shows the 'Network Settings' window. At the top is a menu bar with buttons for 'System', 'Preferences', 'Hardware', 'Network', 'Accept', and 'Quit', each with a 'File' sub-button. Below the menu bar is the 'Network Settings' panel. It contains two main sections: 'Gateway Parameters' and 'Advanced Options'. The 'Gateway Parameters' section has input fields for 'Suite Name' (ct34), 'Host Name' (bay34), 'IP Address' (3.57.255.34), 'Net Mask' (255.255.252.0), 'Broadcast Address' (3.57.255.255), and 'Default Gateway' (3.57.252.254). The 'Advanced Options' section has a checkbox for 'Advanced Options' which is checked, and a sub-section 'Use NIS?' with an unchecked checkbox. Below this is an 'Internal Option' section with an unchecked checkbox and an 'Internal Subnet' field set to 192.9.220.

Figure 4-4 Networks Settings Screen

- 3.) Enter the hospital provided Host Name.
The Host Name identifies the network hostname and AE Title of the CT system.
The Host Name:
 - **MUST NOT** be <Suite Name>_oc or <SUITE NAME>_oc.
 - **MUST NOT** exceed 16 Characters.
 - **MUST** only contain the following characters: A through Z, a through z, 0 through 9, - and _
- 4.) Enter the hospital provided IP Address.
- 5.) Enter the hospital provided Net Mask (if the CT system is on a subnet).
- 6.) Enter the Broadcast Address:
The Broadcast Address should be the same as the IP Address except for the bits of the host id portion (last digit group) set to 1s or 0s depending on the configuration of the network. The standard default is 1s but older SunOS machines used 0s.

Example: If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the broadcast address.
If the network contains genesis based scanners or other SunOS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.
- 7.) Enter the hospital provided Default Gateway IP Address in the Default Gateway field (if applicable). If the site network does not use a default gateway, leave the field blank.
- 8.) Select NIS (Yellow Pages database) Advanced Option only if requested by the hospital network administrator as follows:
 - a.) Select ADVANCED OPTIONS button on the Network Settings screen.
 - b.) Select Use NIS? button.
 - c.) Enter the hospital provided Domain Name.
- 9.) Record all the CT Network parameters in the *LightSpeed 4.X Software Installation Procedures* Document, or on the worksheet in 2.2 on page 24.

5.3 Initiate System Reconfiguration

- 1.) Select ACCEPT on the LightSpeed 4.X System Configuration Screen.
The system loads the application software, OS patches, and kernel changes, and configures the system on both the OC and the SBC.
This loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a shell window, which closes when the loading process is complete. All the window output is logged to a file named:
`/var/adm/install.log.YYYYMMDDWWHHMMSS.`
(Where YYYYMMDDWWHHMMSS is the Date/Time that the loading process was started.)
- 2.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click on YES.
- 3.) The system will automatically login as ctuser after the reboot. Select OK on the Autostart Disabled popup message.
- 4.) To startup Applications, in the console shell window, type **startup** ENTER.

Section 6.0

Declaring Remote Hosts on the LightSpeed 4.X System

6.1 Enter LightSpeed 4.X Remote Host Configuration Screen

- 1.) On the OC, select the IMAGE WORKS icon.
- 2.) Select NETWORK.



6.2 Declaring Advantage NET Remote Hosts on the LightSpeed

Use Advantage NET Protocol networks to communicate with older CT or MR Systems (MR Signa 4.x, CT-HLA, CT/I Systems, and Workstations that support the Advantage NET protocol). Advantage NET Protocol does not offer full compatibility with LightSpeed DICOM formats.

Repeat the following procedure for each Advantage NET Remote Host device that the customer expects to have this CT system communicating with.

- 1.) Select REMOTE HOSTS from the pull down menu. The system displays the Remote Host Parameter Screen as shown in [Figure 4-5](#).

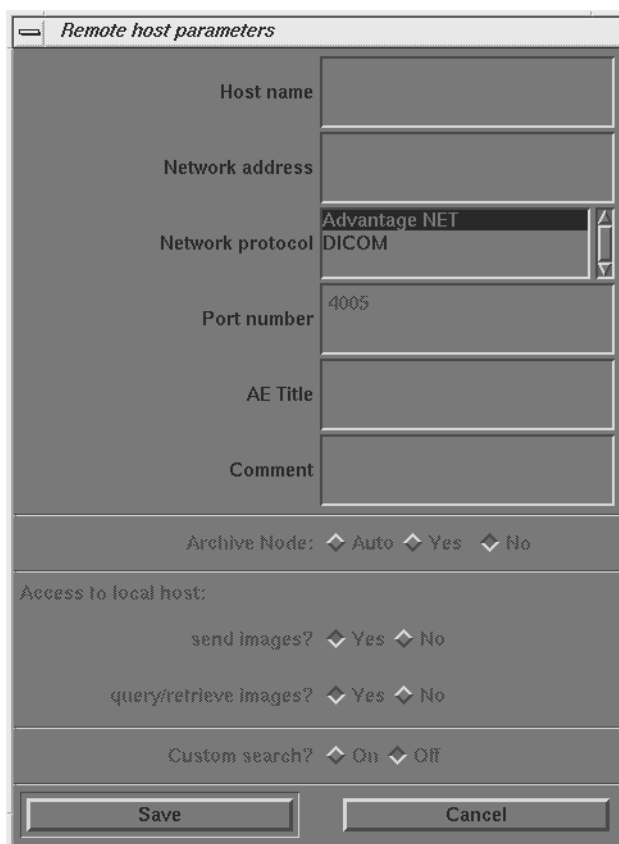


Figure 4-5 Advantage Net Network Protocol Parameter Settings

- 2.) Enter the hospital provided Host name.
- 3.) Enter the hospital provided Network Address (IP Address).

- 4.) Select ADVANTAGE NET as the Network Protocol.
The systems automatically de-highlights the remaining parameter fields on the Remote Host parameter selection screen. These are dedicated DICOM protocol parameters and do not apply to Advantage NET type devices.
- 5.) Select SAVE to store the parameter settings of the remote host.

6.3 Declaring DICOM Remote Hosts on the LightSpeed

Use DICOM protocol networks to communicate to DICOM devices such as CT/i, CT Synergy, DLX, MR Lx, and third party hosts.

Repeat the following procedure for each DICOM remote host device that the customer expects to have this CT system communicating with.

- 1.) Select REMOTE HOSTS from the pull down menu. The system displays the Remote Host Parameter screen as shown in [Figure 4-6](#).

Figure 4-6 DICOM Network Setting Protocol Parameter Settings

- 2.) Enter the hospital provided Host name.
- 3.) Enter the hospital provided Network Address (IP Address).
- 4.) Select DICOM as the Network Protocol.
The system automatically highlights the remaining parameter fields on the Remote Host parameter selection screen. These are dedicated DICOM Protocol parameters.
- 5.) Enter the TCP/IP Listening Port Number from the DICOM Conformance Statement provided with the device.

- 6.) Enter the **AE Title** from the DICOM Conformance Statement provided with the device.
Application Entity Titles (also known as ACR-Nema or Dicom Name) refer to the DICOM Network Services that a device provides to the CT System. For most devices, the AE Title is the same as the hostname (LightSpeed systems are equipped with this feature).
However, some devices such as PACS systems may have separate AE Titles and port numbers for each of the services that the PACS system provides. In these cases, you must enter a separate remote host (same hostname and IP Address) for each of the independent AE Title Services that the host provides (one host as an image **push-to** destination, another host as a **query/retrieve** provider, and another host as a **storage/commitment** provider).
Be sure to review the DICOM Conformance Statement for each device that will provide a remote host network service for the CT system (image **push-to** or store destination, Query/Retrieve, and Storage Commitment) to ensure that each service is correctly configured.
- 7.) Select the correct **Archive Node** choice for the device. The Archive Node selection field defines the ability of the remote host to act as a DICOM Storage/Commitment provider and indicate to the operator that a study/series/image was archived. Select:
 - **AUTO** to have the CT system automatically check to see if the designated remote host is a DICOM Storage/Commitment Provider.
 - **YES** if the device is the hospital designated DICOM Storage/Commitment Provider. During an Application Study Archive process, the local browser screen will indicate **Archive Status = Y** to the operator.
 - **NO** if the device is not a DICOM Storage/Commitment Provider.
- 8.) Select the correct **Access to local host:** settings. These two selections allow you to selectively block the remote host from using the LightSpeed DICOM services as a provider (image **push-to** destination, and a **Query/Retrieve** provider).
 - **Send Images?** Set to **YES** if the customer wants the CT system to be able to have images **pushed** to the system from the applicable remote host. Set to **NO** if the customer wants to block an image **push** from the applicable remote host.
 - **Query/retrieve images?** Set to **YES** if the customer wants the remote host to be able to review the CT image database (query) and **pull** selected images from the database. Set to **NO** if the customer does not want the remote host to have this ability.
- 9.) Select the correct **Custom search?** setting. This selection allows the CT scanner to selectively search through the remote host's image database when the operator is using remote browser screen to **query** the remote host. The search parameters that the LightSpeed system allows the customer to use are: last name contains, patient ID, exam number, accession number, and exam date.
 - Select **ON** if the device supports custom searches as part of the devices **Query/Retrieve** DICOM Provider service.
 - Select **OFF** if the device does not support custom searches.
- 10.) Record all the LightSpeed remote host network parameters for each remote host in the *Software Installation Procedures* manual.
- 11.) Select **SAVE** to store the parameter settings of the remote host.

Section 7.0

Declaring the CT System on Remote Hosts

7.1 Declaring LightSpeed on Advantage NET Protocol Devices/Systems

Refer to the appropriate service manual provided with the Advantage NET Protocol device or system to find instructions how to declare the CT System as an Advantage NET remote host.

7.2 Declaring LightSpeed on DICOM Protocol Devices/Systems

Refer to the appropriate Service Manual provided with the DICOM protocol device or system to find instructions how to declare the CT System as a DICOM remote host.

The CT System provides two DICOM Services as a provider to remote hosts:

- A remote host can **push** images to the CT image database.
- A remote host can review the CT image database (query) and **pull** selected images (retrieve).

Use the following LightSpeed parameter information to configure the DICOM device/system to either **push** images to the CT scanner and/or perform a **Query/Retrieve** operation:

- **Hostname:** Provided by the Hospital Network Administrator. Exactly the same LightSpeed assigned hostname entered in Network Configuration Screen.
- **Application Entity Title:** Exactly the same entry as the Hostname.
- **Network Address:** Provided by the Hospital Network Administrator. Exactly the same LightSpeed assigned IP Address entered in Network Configuration Screen.
- **Network Protocol:** DICOM 3.0.
- **Port Number:** For all DICOM service that the LightSpeed System provides, use 4006.
- **Provider Type:** This field concerns the LightSpeed DICOM **Query/Retrieve** provider capability. All LightSpeed systems are wstudy root systems, which allow queries at the exam, series, and image level.
- **Support Worklist:** This field concerns whether a DICOM **Query/Retrieve** provider capable device or system supports a filter search of the image database. All LightSpeed systems support a filtered search of the image database as part of the LightSpeed DICOM **Query/Retrieve** provider capability.

Section 8.0 DICOM HIS/RIS Setup

8.1 Prerequisites

Most hospital HIS/RIS systems are not DICOM compatible and require a DICOM HIS/RIS Worklist Interface to provide patient scheduling information to the CT system. Contact your local HNS support engineer to determine exactly what DICOM HIS/RIS Interface is appropriate for the customer.

In addition, the CT system must have the ConnectPRO software option installed to utilize the DICOM Protocol Worklist capability.

8.2 Loading ConnectPRO Software Option on the CT System

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon. 
- 2.) Click on the UTILITIES icon. 
- 3.) Click on INSTALL.
- 4.) Select INSTALL OPTIONS and click START. The console displays the Software Options window. (Refer to the appropriate *Software Installation Procedures* manual, for screen details.)
- 5.) Insert the ConnectPRO Options disk (DVD) into the drive.
- 6.) Click on OK to continue.
- 7.) On the Software Option window, select CONNECTPRO in the Available Options window.
- 8.) Click on the INSTALL button.

The console may display a message box during the software loading operation. When the system has completed loading the software:

- the Installed Options window displays the ConnectPRO option, and
- the console displays the ConnectPro Setup window (as shown in [Figure 4-7](#))

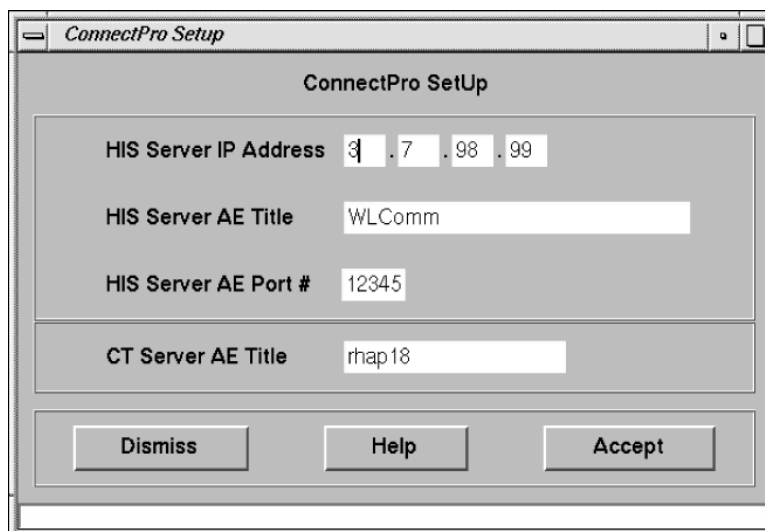


Figure 4-7 ConnectPro Setup Screen

- 9.) Enter the hospital provided HIS Server IP Address.
- 10.) Enter the HIS Server AE Title from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 11.) Enter the HIS Server AE Port # from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 12.) Record all the LightSpeed ConnectPRO HIS/RIS Network parameters in the appropriate *Software Installation Procedures* document.
The system will automatically load the LightSpeed CT Server AE Title (Hostname of the system).
- 13.) Click on the ACCEPT button. The system will check valid setup entries, execute a perl script that makes changes to the configuration files for HIS/RIS, and return to the Software Option window.
- 14.) Click on the QUIT button and the subsequent OK button in the message window.
- 15.) Remove the options disk from the drive, and if using an MOD, write protect the side containing the ConnectPRO option.

Section 9.0

DICOM Filming Devices Setup

9.1 Prerequisites

Before configuring DICOM filming devices (cameras, printers) on the CT System, ensure the following are complete:

- Filming Device Service Representative to assist in camera/printer setup for best CT image quality film presentation.
- Hospital DICOM network is operational.
- Filming device is connected to the DICOM network with the correct filmer DICOM interface.
- Filming device is DICOM protocol compatible.
- Filming device DICOM Conformance Statement document is available.

Note: **Filmer DICOM Application Entity Titles may be site specific. Make sure that you check with the Filming Device Service Representative and the hospital network administrator to ensure you are using the correct AE Title for the destination filming device.**

9.2 Declaring DICOM Filming Devices on the CT System



NOTICE Potential For Data Loss

Empty all filming queues before modifying camera parameters.

This section contains procedures for recording important Camera setup information. Use the table(s) at the end of this section to record information from the setup screens.

- 1.) Click on the SERVICE DESKTOP button.
- 2.) On the Desktop Toolbar select UTILITIES -> INSTALL -> INSTALL CAMERA. The LightSpeed 4.X Install Camera window appears, along with a warning message pop-up box, to remind you that all filming queues must be empty before you begin to update or delete a camera.
- 3.) The Graphical User Interface displayed shows a list of cameras installed (See Figure 4-9). First, you must click OK in the warning message box. See Figure 4-8.

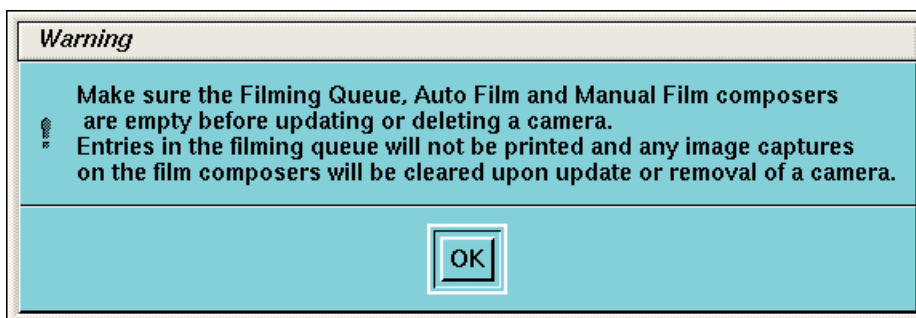


Figure 4-8 Warning Screen

- 4.) Now you are asked a series of questions.
- a.) To add a new camera, click the ADD button (See Figure 4-9).

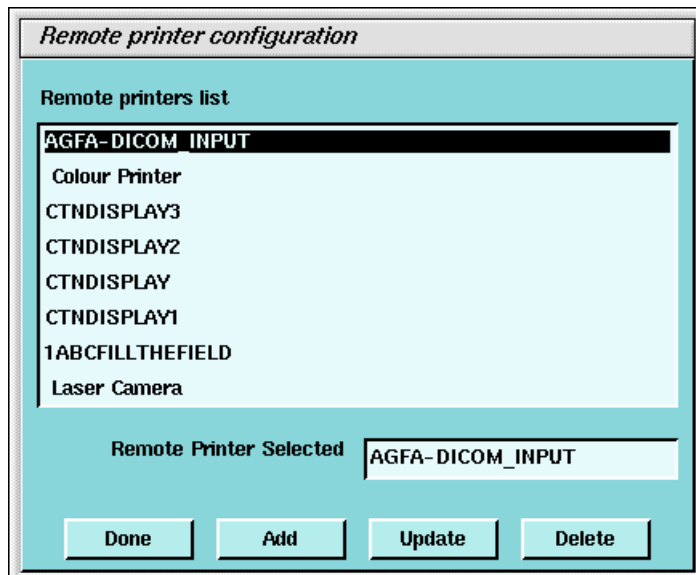


Figure 4-9 Printer Configuration GUI

- b.) A dialog window for the camera type (DASM/DICOM) appears. If no DASM is detected during the OC boot, the DASM button will be disabled (Figure 4-10). If a DASM is present and has not been detected, reboot the OC and run the camera configuration tool again.

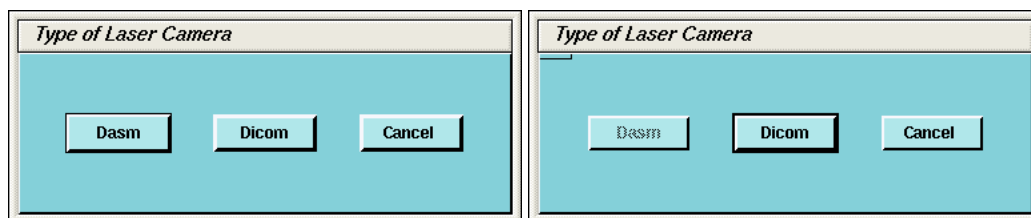


Figure 4-10 Dialog Box for camera Type (with and without DASM)

- 5.) To add a new laser camera, click DASM in the camera type dialog box. This brings up a list of available camera models. Select the appropriate model from the list and click SELECT (See Figure 4-11). Now you must configure it.

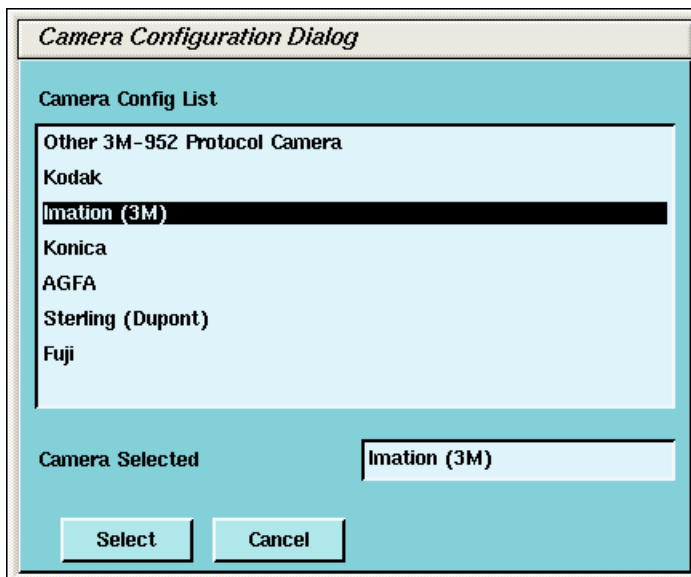


Figure 4-11 Camera Model Dialog (with DASM)

- a.) DASM Interface is automatically detected as either Analogue or Digital
- b.) Two Laser Options are available for laser cameras: SLIDES and ZOOM. Set this option only if the camera being installed supports slides and zoom. Setting the option allows it to be enabled or disabled at the application level.
- c.) Camera manufactures provide two (2) Magnification Type options for cameras. The SMOOTH resolution blurs the image, while the SHARP resolution makes the image pixels more pronounced. The default is smooth.

Comment:

To film good images, and have them look like images filmed by other GE CT products, the following camera settings are suggested:

Kodak: SMOOTH

Dupont/Sterling: SMOOTH

3M/Imation (Laser Camera): SHARP

3M/Imation (Dry View): SMOOTH

Agfa: SMOOTH

- d.) Select the appropriate File Format. Select ON from the drop down list boxes on the menu. Valid film formats are determined by the camera manufacture. IMATION for example, doesn't support 4x4, 2x4 or 1x2 and AGFA does not support 2x4) The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4).

When finished setting parameters, click on DONE and proceed to step 8.

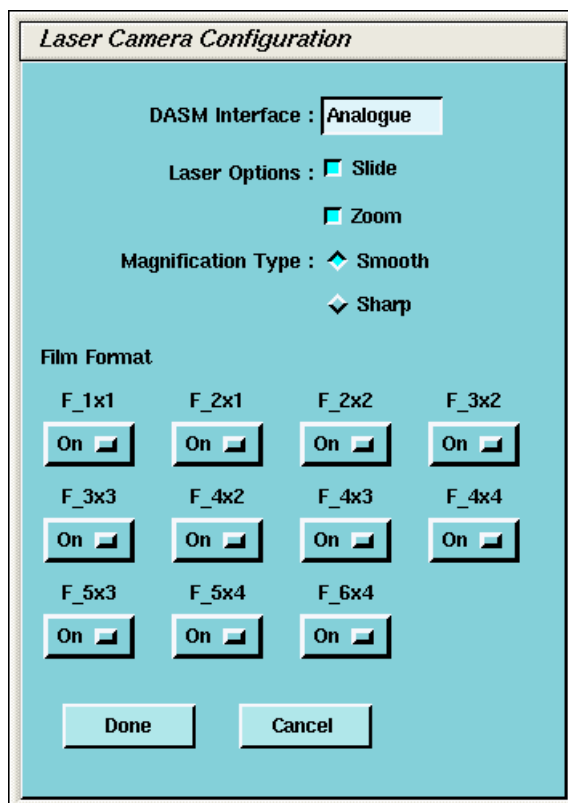


Figure 4-12 Laser Camera Configuration

- 6.) To add a new DICOM camera, click on ADD and then DICOM in the dialog box that appears.
 - a.) A list of camera models appears (See Figure 4-13). Select the appropriate model from the list and click SELECT. Clicking SELECT presets all the parameters to that models except the Network parameters.

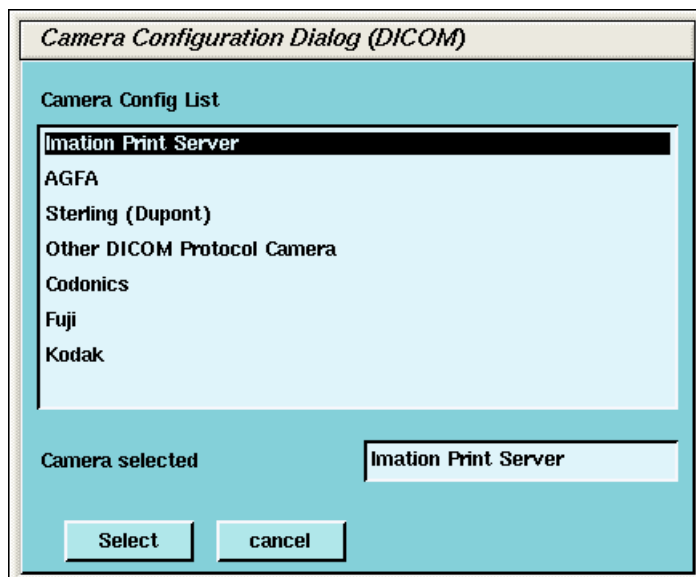


Figure 4-13 Camera Model Dialog (DICOM)

*Comment:
It's advised to
recheck the
preset
information with
the camera
manufacturer's
representative.*

- Selection of a different camera model clears the Image Quality parameters, because these are camera manufacture dependent.
- b.) Enter the Network Parameters (See Figure 4-14)
- > Device Name A unique name used to identify the camera.
 - > Host Name DICOM print server host name, as defined by the hospital.
 - > IP Address DICOM print server IP address, as defined by the hospital.
 - > AE Title DICOM print server application entity title, as defined by the print server. *You should consult the manufacturers DICOM Conformance Statement.*

Note: The Application Entity Title for the Camera may be site specific. Make sure that you check with the Camera Manufacturer's Representative and the hospital network administrator to ensure you are using the correct AE Title for the destination DICOM Print Camera.

- > TCP/IP Listen Port DICOM print server TCP/IP listen port, as defined by the server. *You should consult the manufacturers DICOM Conformance Statement.*
- > Comments Optional comments used by the DICOM print server.

DICOM printer configuration parameters

Device Name:

Host Name:

IP address:

Application title:

TCP listen port:

Comments:

Medium Type:

Destination:

Film orientation:

Magnification type:

Standard film formats:

F_1x1 On	F_2x1 On	F_2x2 On	F_3x2 On
F_3x3 On	F_4x2 On	F_4x3 On	F_4x4 On
F_5x3 On	F_5x4 On	F_6x4 On	

Figure 4-14 DICOM Camera Configuration

- c.) Medium Type specifies the type of film being used. Currently, only BLUE FILM and

CLEAR FILM are supported.

- d.) Set Destination to the final location for film output: either MAGAZINE or PROCESSOR.
- e.) Orientation selects film orientation. Only PORTRAIT is currently supported.
- f.) Set the Magnification Type. This parameter selects the algorithm used to interpolate pixels for proper film resolution. Set this parameter after consulting the camera manufacture to ensure optimal image quality. Choices are describe below:
 - > None No interpolation. This option is not supported by all cameras.
 - > Replicate Adjacent pixels are interpolated. This can result in "pixelized" images. *This algorithm is not normal preferred.*
 - > Bilinear A 1st order interpolation of pixels. Results in images usaually described as blurred. *This algorithm is not normal preferred.*
 - > Cubic A 3rd order interpolation. Used with a large number of possible formulations. Camera manufactures define parameters called "smoothing type" to set coefficients used in this algorithm. The implementation of these "smoothing soefficients" is camera dependent.
- g.) Select the Standard Film Formats. Select the film format by choosing ON in the pull-down menu box located below each selection. See Figure 4-14. Valid film formats are set by the camera manufacture. IMATION for example, doesn't support 4x4, 2x4 or 1x2 and AGFA does not support 2x4) The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4).
- h.) After the camera has been configured, click the ADVANCED button. This creates the camera device file for you automatically and pops up the Advanced Parameters screen. See Figure 4-15.

Comment: For most Camera Manufacturers, the preferred selection is CUBIC.

The screenshot shows a dialog box titled "Camera Attribute Dialog". It contains several sections of controls:

- Image Quality Parameters** (Section Header)
- Configuration**: A text box containing "PERCEPTION_LUT=KANAMC".
- Maximum Density**: A text box containing "320".
- Minimum Density**: A text box containing "32".
- Smoothing Type**: A text box containing "140".
- Empty Density**: A text box containing "Black".
- Border Density**: A text box containing "Black".
- Film Size**: A text box containing "<empty>".
- Trim**: A text box containing "Yes".
- Priority**: A text box containing "High".
- Buttons**: "Done" and "cancel" buttons at the bottom.

Figure 4-15 Advanced Parameters (Camera Attribute Dialog)

- 7.) Advanced camera parameters control the image quality of films.

- Note: For more information on the proper settings for these parameters, please see the Camera's DICOM Print Device Conformance Statement or the Camera Manufacturer Representative. You may need to refer to a copy of the Conformance Statement as you are working with the Camera Manufacturer's Representative, to correctly set up the DICOM Print Camera I/Q and Time-out Settings.
- a.) **Configuration** - This parameter is camera manufacturer dependent as is typically used to specify the image contrast. The Configuration field may be up to 1024 characters long. The field will scroll automatically as text is entered. To review your entry, simply click and hold the middle mouse button, while the cursor is in the field, and drag the mouse towards the right (or left) as needed.
- Note: Recommended Configuration Setting Values:
Agfa Drystar (MG3000) **PERCEPTION_LUT=KANAMORI (100)**
Imation Dryview (8700) **LUT=0,7**
Kodak Laser Printer 190 **CS434\CN0\PD1.20**
- b.) **Smoothing Type** - Set Smoothing Type to ON, and input the selected value. This parameter is used when the magnification type is CUBIC. It represents the coefficient for the image resolution algorithm. This parameter is camera manufacturer dependent, and should be re-verified with your radiology department.
- Note: **Recommended Smoothing Type Starting Values and Ranges:**
Agfa DryStar (MG3000) Start Value: **140** Range: **137-150**
Imation Dryview (8700) Start Value: **3** Range: **3-13**
Kodak Laser Printer 190 Start Value: **Enhanced** Range: **Normal**
- c.) **Minimum and Maximum Density** - Used to set brightness of the images on film. The range of values is 0-4095, although the valid range for a specific camera is manufacture dependent. For **Maximum Density**, input the correct value into the text box. For **Minimum Density**, set it to ON and input the correct value in the text box.
- Note: **Recommended Minimum and Maximum Density Starting Values:**
Agfa Drystar (MG3000) Min.: **20** or **23** Max: **300**
Imation Dryview (8700) Min.: **(Blank)** Max: **300**
Kodak Laser Printer 190 Min.: **20** Max: **300**
- d.) **Empty Image Density** - This parameter sets the density for empty film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.
- e.) **Border Density** - This sets the density for the border used around the film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.
- f.) **Film Size** - Allows the system to specify a particular film size, if selected.
- g.) **Trim** - YES produces a white (clear) box surrounding each image.
- h.) **Priority** - This sets the print priority.
- i.) If you have completed entry of advanced parameters, click DONE.

CAMERA DATA TABLES: TO LOCATE INSTALL CAMERA INFORMATION: CLICK ON THE SERVICE DESKTOP BUTTON. ON THE DESKTOP TOOLBAR SELECT UTILITIES -> INSTALL -> INSTALL CAMERA. THE LIGHTSPEED 4.X INSTALL CAMERA WINDOW APPEARS. SELECT EACH OF THE CAMERAS THAT ARE INSTALLED FROM THE LIST OF INSTALLED CAMERAS, AND CLICK ON UPDATE TO VIEW THE CAMERA'S SETTINGS. RECORD THE VALUES USED TO SET UP EACH CAMERA IN THE TABLES THAT FOLLOW.

EXTRA TABLES ARE PROVIDED FOR MULTIPLE CAMERAS

Note: You can determine this information by looking at the contents of the following files:

- For a DASM Camera: `/usr/g/ctuser/app-defaults/devices/camera.dev`
- For a DICOM Print Camera: `/usr/g/ctuser/app-defaults/devices/name.cfg`
where, *name.cfg* is the camera device name from the printer configuration GUI.

Example: more <filename from above> ENTER

DASM CAMERA #1 SETUP

GUI SETTING	SELECTIONS	VALUE
Camera Type	Model Type of Camera	
DASM Type	Digital or Analog	☐ Digital ☐ Analog
Options	Slides or Zoom	☐ Slides ☐ Analog
Film	Smooth or Sharp	☐ Smooth ☐ Sharp
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Film Format Default	1x1, 2x1, 2x2, 3x2, etc.	

Table 4-1

DICOM CAMERA #1

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 4-2 DICOM Camera #1

GUI SETTING	SELECTIONS	VALUES
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 4-2 DICOM Camera #1

DICOM CAMERA #2

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 4-3 DICOM Camera #2

GUI SETTING	SELECTIONS	VALUES
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 4-3 DICOM Camera #2

DICOM CAMERA #3

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 4-4 DICOM Camera #3

9.3 Troubleshooting Tips

The following is a summary of troubleshooting information for DICOM print that was gathered during software testing and validation of the DICOM print feature.

There is also a significant amount of additional troubleshooting procedures, and the theory of the DICOM print feature in the System Service Manual. Should you have problems installing a DICOM print camera, first read the information in System Service Manual.

ERROR BRINGING UP THE CAMERA INSTALLATION/FILMING APPLICATION

- **Symptom:** After creating/modifying the DICOM print camera the install camera interface does not come up and the filming application indicates it cannot bring up the film composer.
Cause: The configuration field contains invalid information such as a backslash (\) as the final character in the field or brackets ({}).
- **Solution:** The camera.dev file in `~ctuser/app-defaults/devices` must be manually edited to remove the offending characters in the set configuration line. Invalid characters include \{ }

CONFIGURATION INFORMATION FIELD

- **Symptom:** Cannot view the entire configuration field (> 25 characters)
- **Solution:** Hold down the middle mouse button and move the field contents

NEED TO SET DICOM PRINT ATTRIBUTES NOT SUPPORTED BY SOFTWARE

- **Symptom:** User wants the white border around each image box ON/OFF permanently for this system and it cannot be set as the default for the camera.
- **Solution:** Using your favorite editor, add the following line to the camera.dev file located in `~ctuser/app-defaults/devices` after the DICOM print device has been otherwise configured.

For Trim Off - set	TRIM NO
For Trim On - set	TRIM YES
- **Symptom:** DICOM print camera supports multiple film sizes and the user only wants to print if the film size is correct for LightSpeed 4.X (14x17). [Otherwise the camera will queue the films or return an error causing the queue to pause (based upon the DICOM print camera specifications).]
- **Solution:** Using your favorite editor, add the following line to the camera.dev file located in `~ctuser/app-defaults/devices` after the DICOM print device has been otherwise configured.

To force a 14x17 film size - set	filmSize 14INX17IN
----------------------------------	--------------------

NEED TO PREVENT DICOM PRINT ATTRIBUTES FROM BEING SENT TO DICOM PRINT CAMERA

- **Symptom:** Some DICOM print attributes are optional, and may result in fatal errors. For example, the Fuji camera does not support the Empty Image Density parameter for the film box.
- **Solution:** Using your favorite editor, add the following line(s) to the camera.dev file located in `~ctuser/app-defaults/devices` after the DICOM print device has been otherwise configured.

- | | |
|--|-----------------|
| - To prevent sending the Smoothing Parameter set | FB_Smooth FALSE |
| - To prevent sending the Border Density set | FB_Border FALSE |
| - To prevent sending the Empty Image Density set | FB_EID FALSE |
| - To prevent sending the Minimum Density set | FB_MinD FALSE |
| - To prevent sending the Trim Parameter set | FB_Trim FALSE |

ERROR TRYING TO CONNECT TO THE DICOM PRINT CAMERA

- **Symptom:** DICOM print server can be reached (ping), but Application error indicates "Unable to start filming interface" and the help message talks about running the install.dasm (Association Error)
- **Solution:** The system is unable to complete the association. Check the `AE Title` and the `Port number` of the DICOM print server and correct them through the LightSpeed 4.X Install Camera procedure.

FILM COMPOSER ERROR NOT USABLE

- **Symptom:** Film composer error says "unrecognized status - code 0"
- **Solution:** Review the log file, the attention and status windows. These areas have the correct filming status (for example, **film jam** and **supply empty**).

DEBUGGING CONNECTION ISSUES DIFFICULT

- **Symptom:** The timeouts for the DICOM print are very long, which means one needs to wait a long time before you know the application is not working.
- **Solution:** The timeouts for the DICOM print were setup to ensure that the system would work regardless of whether the DICOM print camera was on a LAN or a WAN halfway around the world. The DICOM print timeouts for the association and DIMSE classes (for example, **N-GET**, **N-DELETE**) can be modified within the DICOM print camera installation. They can be reduced down to 90 seconds.

DICOM PRINT ERROR ON N-GET TIMEOUT CONFUSING

- **Symptom:** When the N-GET timeout goes off, the error message in the prslog file will be "Could not get printer status, invalid command sequence for N-GET".
- **Solution:** When the user sees the above error they may want to consider that the issue may be an inactivity timer on the N-GET DIMSE service.

DICOM PRINT CAMERA SLIDE SUPPORT

- **Symptom:** Current implementation of DICOM print does not allow selection of slide format.
- **Solution:** Feature not currently supported. Possibly in future releases.

CONFUSION ON FILM FORMAT NOTATION

- **Symptom:** GEMS Laser Camera and DICOM Print film format notations are opposite.
- **Solution:**
 - GEMS Laser Camera film format notation has always been row x col (for example, 12 on 1 = 4x3)
 - DICOM Print Standard film format notation is col x row (for example, 12 on 1 = 3x4)

Section 10.0

Teleradiology (Framegrabber Type) Systems & Aux. Monitors

10.1 Introduction

Some Genesis based systems have teleradiology (TR) systems that framegrab the Genesis GFB video (512 x 512 50/60Hz). LightSpeed 4.X DOES NOT directly support this type of TR. The LightSpeed 4.X RGB color display video is a much larger format at a much higher pixel frequency. GE does not promise any direct compatibility with framegrabbing TR systems (DICOM 3.0 TR systems may work depending on the DICOM implementation but GEMS does not and cannot validate all the various TRs.)

In the framegrabber case, a high quality (300Mhz bandwidth) video splitter/amplifier (as listed above) is needed to intercept and re-drive the display CRT RGB video. Composite greyscale would then be available on Green #2 (1280x1024 pixels at 72Hz). Any framegrabber hardware attempting to capture this signal must be capable of a 140Mhz pixel rate. This also involves TR system configuration parameters. The TR capture software may also need upgrading to deal with 1280x1024 and/or **crop** the signal. The TR remote display software may need upgrading to view the larger format. The image transmission times to the remote TR may be up to 4 times as long. GEMS will supply all technical information necessary to assist TR suppliers in making their systems work with LightSpeed 4.X, but GEMS cannot be responsible for this third party TR equipment, software, or compatibility with LightSpeed 4.X.

The following common parts are available from GE Medical Systems:

PART NUMBER	DESCRIPTION
2237018-2	Four-way Video Splitter (BNC Converter)
2154425	BNC to BNC 7' Splitter Cable
2142221	BNC to Octane 7' video Cable
2256482	DB15 to Octane 7' Video Cable
2256485	DB15 to BNC 7' Adapter Cable

Table 4-5 Common Parts Available From GEMs

10.2 Auxiliary Monitors Setup

Figure 4-16 illustrates the **only approved GE CT configuration** for remote monitors, including the part numbers for the only approved components. The use of other configurations and/or components should be strictly avoided.

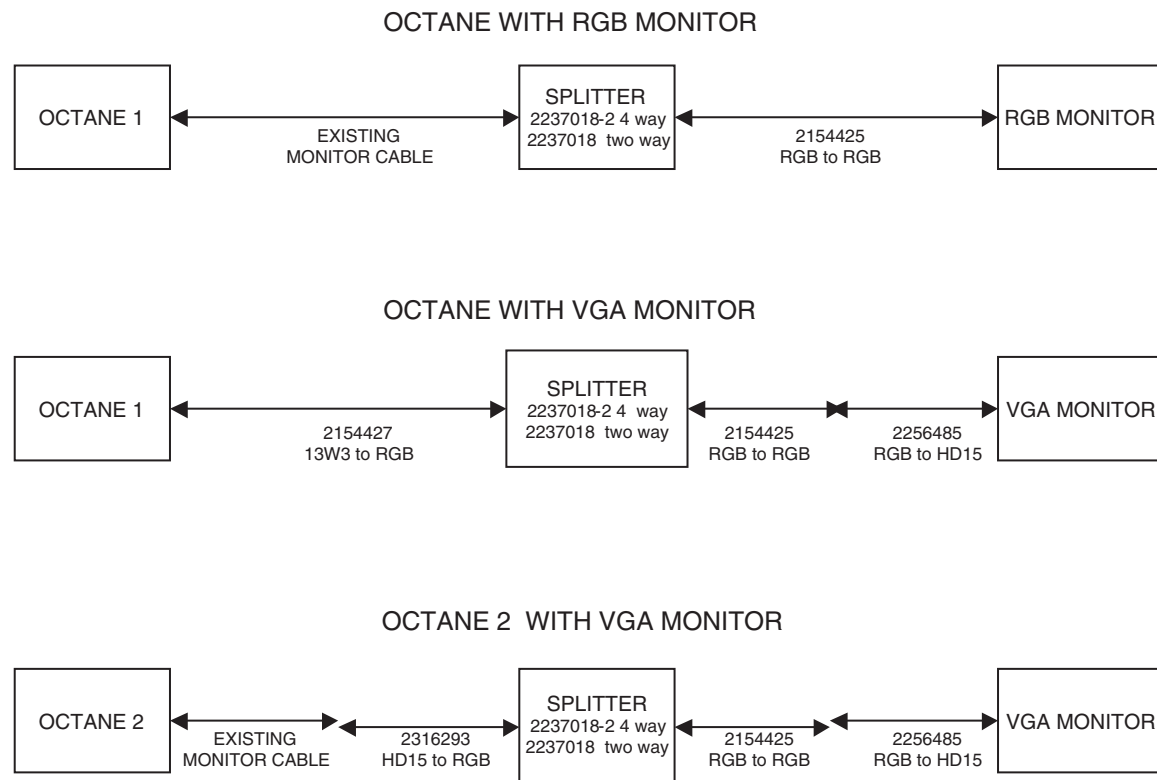


Figure 4-16 Video Splitter / Monitor Cabling

10.3 GEMS LightSpeed 4.X Analog Filming Interface Specs (Video & Serial)

Note: The analog (VDB) video output specs and serial interface were specified from, and are the same as, the Genesis interface output specs. This does NOT apply to the LightSpeed 4.X digital (LCAM) filming interface.

OUTPUT	VALUE
amplitude	1 volt peak-to-peak
video	0.643V $\pm$ 10%
setup	0.071V $\pm$ 10%
sync	0.286V $\pm$ 10%
DAC resolution	8 bits
diff. linearity	$\pm$ 1 LSB max
glitch area	80 picovolt seconds max, for any step size
rise/fall times	> 10 nsec, $\pm$ 10-90%
FS settling time	7.5 nsec typical to 1 LSB
transfer function	guaranteed monotonic
noise level	> 5.0 millivolt peak-t~peak, combined sync/async noise
DC offset	$\pm$ 1 VDC referenced to ground

Table 4-6 Video Output (Measured Into 75Q at BNC Output)

OUTPUT	VALUE
logic family	F series TTL
output low level	0.8VDC max
output high level	2.0VDC min
output period	41.336 nsecs $\pm$ 10%
transition times	10 nsec max, $\pm$ 10-90%

Table 4-7 Pixel Clock Output

VIDEO TIMING CHARACTERISTIC	60HZ	50 HZ
pixel frequency	24.192 Mhz	24.192 Mhz
pixel period	41.336 nsec	41.336 nsec
horizontal line frequency	33.6 Khz	33.6 khz
horizontal line width	720 pixels	720 pixels
horizontal active	544 pixels	544 pixels
horizontal blanking	176 pixels	176 pixels
horizontal front porch	26 pixels	26 pixels
horizontal sync	76 pixels	76 pixels
horizontal back porch	74 pixels	74 pixels

Table 4-8 Video Timing Characteristics

VIDEO TIMING CHARACTERISTIC	60HZ	50 HZ
vertical frame frequency	60Hz	50Hz
vertical frame time	560 lines	672 lines
vertical active	524 lines	524 lines
vertical blanking	36 lines	148 lines
vertical sync	3 lines	3 lines
vertical back porch	30 lines	86 lines
vertical front porch	3 lines	59 lines
scanning format	non-interlaced	non-interlaced

Table 4-8 Video Timing Characteristics (Continued)

FORMAT	VALUE
visible field	544 pixels by 524 lines
image field	512 pixels by 512 lines
greyscale field	32 pixels by 16 level grey bar on left side of image
greyscale	software selectable on/off
greyscale off value	0 (black)
initial greyscale	255 (white) at upper left corner
border field	12 lines at bottom of visible field
border field value	any 8bit value, software programmable

Table 4-9 Video Display Format

ATTRIBUTE	VALUE
interface	R5422
25D conn pinout	pin 8 (RX+), pin 21 (RX-), pin 9 (TX+), pin 22 (TX-), pin 7 (GND)
baud rate	1200 baud
word length	8 bit, 1 start bit, 1 stop bit
parity	even
type	asynchronous

Table 4-10 Host Communications/Control Serial Port (analog interface only)

Note: The LightSpeed 4.X digital DASM/LCAM serial control has standard R5232 on pins 2, 3, & 7.

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Chapter 5

HHS Forms & Data Verification

Section 1.0

FORM 4879 (Data Record for LightSpeed 4.X)

Note: Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 2.0

Site History

IMPORTANT

- Locate the HHS data, sent from the factory, in the software collector box.
- Retain this data with the HHS data collected at the site, per your local procedure.
- A complete generator RECAL is not required.

Note: If any one of the calibration procedures fail, data collected is different, or you replace any component in the High Voltage section, YOU MUST RECALIBRATE THE SYSTEM. To recalibrate the system, refer to procedures in the System Service manual.

COMPONENT TRACEABILITY

- Complete the Product Locator Report and mail it.
- Add a copy of the Product Locator Data sheet to this chapter of the book.
- Collect all factory supplied HHS Data sheets, all HHS data and all Product Locator information. Retain this information, per your local procedure.

SYSTEM HISTORY

Where Installed _____

City _____ State _____

Customer's Telephone Number _____ Room No. _____

System I.D. No. _____

Operator's Console (Master Control)

Model Number _____ Serial Number _____

System Gantry

Model Number _____ Serial Number _____

Dates of Tests _____ Performed by _____

Table 5-1 Data Record

TEST EQUIPMENT HISTORY

Test Equipment Used (If Applicable)				
Name	Manufacturer	Model Number	Serial Number	Calib. Due
DMM				

Table 5-2 Test Equipment Used

Section 3.0 System Tests

3.1 Generator Calibration

3.1.1 Introduction

3.1.1.1 Process Overview

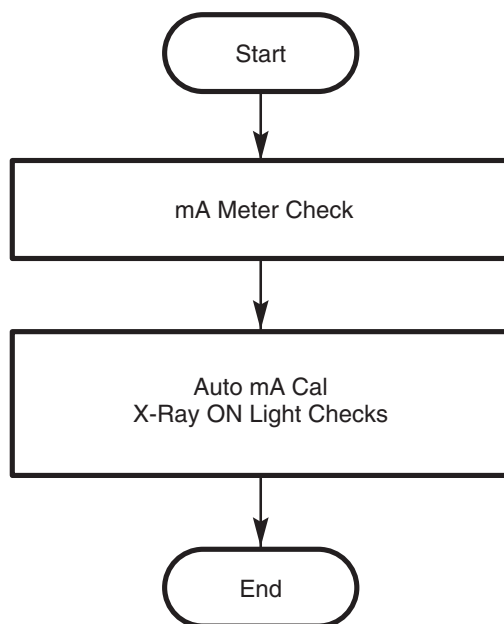


Figure 5-1 Calibration Process Flowchart

3.1.1.2 Required Tool

DVM with meter leads

3.1.2 mA Meter Check with Verification



- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon, .
- 2.) Click on the UTILITIES icon, .
- 3.) Select INSTALL.
- 4.) Select GENERATOR CALIBRATION.
- 5.) Switch OFF the HVDC (550) ENABLE inside the gantry.
- 6.) Switch OFF the AXIAL DRIVE inside the gantry.
- 7.) Rotate the anode tank to the two o'clock position.
- 8.) Pin the gantry.
- 9.) Remove the OBC cover (if necessary).
- 10.) Use a DVM as a DC mA meter. Set the meter range to one decimal place.
- 11.) Connect the DVM black lead to TP8 (ACAL1) on the mA board.
- 12.) Connect the DVM red lead to TP11 (ACAL2) on the mA board.

Collimator
HEMRC
MA Control
KV Control
Gentry I/O
LSCOM
Heurikon (CPU)

Figure 5-2 OBC Board Slot Assignment

- 13.) Select MA METER CHECK.
 - 14.) Select ACCEPT.
 - 15.) Record the displayed and measured Anode mA values in the Circuit OFF and Circuit ON table in space provided below. Acceptable Circuit ON values: 140 to 200mA.
- Note: **The cathode side reads approximately 19 mA during Circuit ON. The anode should read 140 to 200mA, as stated above.**
- 16.) Repeat the test for the cathode side:
 - a.) Use a DVM as a DC mA meter; set the meter range to one decimal place.
 - b.) Connect the DVM red lead to TP9 (CCAL1) on the mA board.
 - c.) Connect the DVM black lead to TP14 (CCAL2) on the mA board.
 - 17.) Select ACCEPT.
 - 18.) Record the displayed and measured cathode mA values in the Circuit OFF and Circuit ON table in space provided below. Acceptable Circuit ON values: 140 to 200mA.
- Note: **The anode side reads approximately 19 mA during Circuit ON. The cathode should read 140 to 200mA, as stated above.**
- 19.) Disconnect the test equipment.
 - 20.) Switch ON the HVDC (550) enable.

3.1.3 mA Meter Verification - HHS Requirement

Circuit ON		
	A/D	Range
Anode mA		15 - 25
Cathode mA		15 - 25
DVM - A/D = Delta Compare Delta to Limits		

Table 5-3 mA Meter Verification (Table 1)

3.1.4 Auto mA Cal

Use this procedure to verify and complete Auto mA Cal on the system. If you have just run a Heat Soak and Seasoning, wait 30 minutes before you run the Auto mA Cal.

Note: **The system automatically warms up the tube.**

- 1.) Select AUTO MA CAL.
- 2.) Press START SCAN when it flashes. The program automatically runs Ductility and Auto mA Cal.
- 3.) During this test, verify the X-Ray on indicators, located on the gantry display and the rear gantry shroud, illuminate during X-Ray on.

If the system cannot complete the Auto mA Cal because HV aborts, run the "Tube Heat Soak and Seasoning" scan protocol (see the *LightSpeed 4.X System Service Manual* for details). Place a 48cm phantom in the X-Ray beam to protect the detector from radiation damage during this procedure. Do not worry about the image quality of the scans reconstructed during the Tube Heat Soak and Seasoning.

☐ Check box when complete Initial: _____

The MX200 tube uses a metal/ceramic frame, which means some electrons hit the target and actually bounce off, and are then captured by the frame. This causes a frame current, which results in an anode mA measurement of slightly less than the cathode mA, typically 5 to 10% less. Because the cathode mA is the true indicator of radiation output, the CT QX/i system controls cathode mA rather than anode mA.

3.1.5 X-ray Lights

- 1.) Perform several scans (see below).
- 2.) Verify that the X-ray ON lights are on during the scans. Check boxes in [Table 5-4](#)

PERFORMING A SCAN

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
- 2.) Select DIAGNOSTICS.
- 3.) Select DIAGNOSTIC DATA COLLECTION
- 4.) Set the scan time to 2.00
- 5.) Set the Kv to 80
- 6.) Set the ma to 40.



Light On?	Light Locations
☐	SCIM
☐	Gantry Front
☐	Gantry Back (use a mirror to view)
☐	Room Light

Table 5-4 X-ray light chart

OPERATOR INDICATORS - HHS REQUIREMENT

Initials: _____

X-Ray Indicators on Front and Back of Gantry Illuminate During X-Ray

3.2 Verify X-ray/Signal to Noise Test

This section uses X-Ray to test the DAS for opens, and shorts.

System Configuration:

All covers ON

Nothing in detector FOV

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
- 2.) Click on the DIAGNOSTICS icon.
- 3.) Select DAS TOOLS.
- 4.) Select INTERCONNECT/X-RAY VERIFICATION.
- 5.) Follow on-screen instructions to acquire each set of X-Ray verification scans--for a total of four sets.
- 6.) Select ACCEPT RX
- 7.) Press START SCAN when it flashes.



- 8.) Select DISMISS to close.
- 9.) Select X-RAY VERIFICATION ANALYSIS.
- 10.) If X-Ray verification scans do not pass, troubleshoot with DAS tools and System Service Manual before continuing.
- 11.) Complete the F4879 Signal/Output check.

Initials: _____ ☐ Check box when complete

3.3 CT Number Check

Refer to [Auto CT N Number Adjustment on page 51](#).

All CT N numbers for all calibrated techniques fall within 0 ± 1.5 HU. All Values 0 ± 1.5 HU? Yes: ☐

3.4 Collimator Aperture Test

3.4.1 Small Aperture Test

- 1.) Clear all obstructions out of the gantry SFOV.
- 2.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon, .
- 3.) Click on the UTILITIES icon. .
- 4.) Select INSTALL.
- 5.) Select COLLIMATOR APERTURE TEST.
- 6.) Select SMALL APERTURE SCAN to display a scan protocol window, and start gantry activity (refer to [Figure 5-3](#)).
- 7.) Press the START SCAN key when it flashes, to initiate the first scan.
- 8.) When the test completes, the screen updates to display the Small Aperture values.

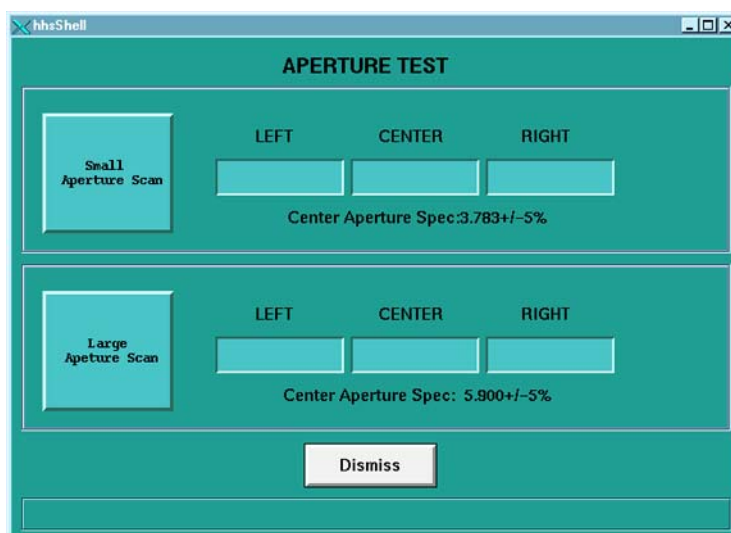


Figure 5-3 Collimator Aperture Test Screen

3.4.2 Large Aperture Test

Note: The center channel is the only channel with a spec. The left and right channels may or may not pass. Small aperture center spec is $3.783 \pm 5\%$. Large aperture center spec is $5.900 \pm 5\%$.

- 1.) Select LARGE APERTURE to display a scan protocol window, and start gantry activity (Refer to [Figure 5-3](#)).
- 2.) Press the **START SCAN** key when it flashes, to initiate the first scan.
- 3.) When the test completes, the screen updates to display the large aperture values.

Record the large and small aperture values in the table below.

COLLIMATOR APERTURE CHECK - HHS REQUIREMENT

Initials: _____ ☐ Check box when complete

Aperture	Center	Spec - Center only
Small		$3.783 \pm 5\%$
Large		$5.9 \pm 5\%$

Table 5-5 Collimator Aperture Check

3.5 Tomographic Plane Indication Accuracy

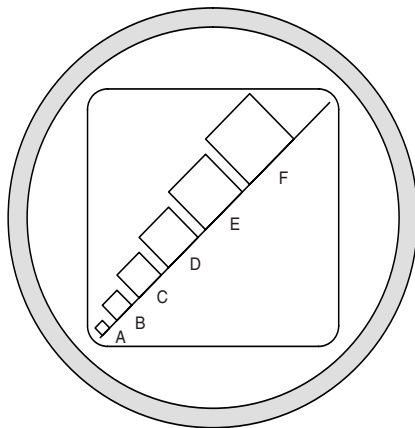
Refer to [Tomographic Plane Indication on page 52](#)

3.6 Image Calibration

☐ check box when complete.

Detailed Calibrations completed and CT N Number adjustment completed. The Number of Phantom Cals vary with site-specific scanning protocols.

UNUSUAL ALIASING, OTHER ARTIFACTS



Visual inspection for unusual aliasing, and other artifacts. Visual results acceptable? Yes: ☐

3.7 Laser Warning Labels

- 1.) Make sure all laser warning labels appear in the correct location on the outside of the gantry.
- 2.) Obtain and install replacements for any missing labels.



Figure 5-4 Laser Warnings and Precautions

Initial: _____

3.8 Operator Use Warning Label - SCIM

Make sure the X-Ray warning label appears in the correct location on the SCIM.

Initial: _____

3.9 Notes on Additional Test Procedures

The following test procedures were completed in manufacturing, prior to shipment. Information on these tests is available on request, and is not required for field records:

- POR
- ISO
- CBF

Chapter 6

Install Completion Checklist

Complete the following checklist:

Section 1.0

CT System

- ☐ Functional tests completed.
- ☐ Power and ground audit completed.
- ☐ Seismic kit installed (if needed in your area).

Section 2.0

CT Options

- ☐ Injector functional tests completed.
- ☐ AWW functional tests completed.
- ☐ Filming/Camera/DASM functional tests completed.
- ☐ UPS functional tests completed.
- ☐ Network items installed and functional tests completed.
- ☐ Verify that all customer software options are installed and functional.
- ☐ Teleradiology Connections (Refer to [Section 10.0 on page 98](#))

Section 3.0

System Cleanup

- ☐ Clean system with supplies shipped with your system.
- ☐ Touch-up nicks and scratches (as needed).
- ☐ Install pads and stickers (as needed).
- ☐ Store customer materials appropriately.
- ☐ Store service documents, tools, and materials appropriately.

Section 4.0

Dollies Return

- ☐ Global Console is shipped on a pallet. Do Not return pallet to GEMS.

Section 5.0

Install Completion Material Checks

- ☐ Complete all HHS documentation. Complete Install CQAs as needed.
- ☐ Complete all dispatching activities (03-04-10 codes).
- ☐ Complete customer acceptance checks/test.
- ☐ Complete system transfer and notify appropriate GE personnel.
- ☐ Communicate with the customer and address installation issues.

Appendix A

Additional Characterization Procedures

Section 1.0 Tilt Characterization

Tools Required: Hex (metric) set
Flat blade screw driver



1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



2.) Click on the UTILITIES icon.

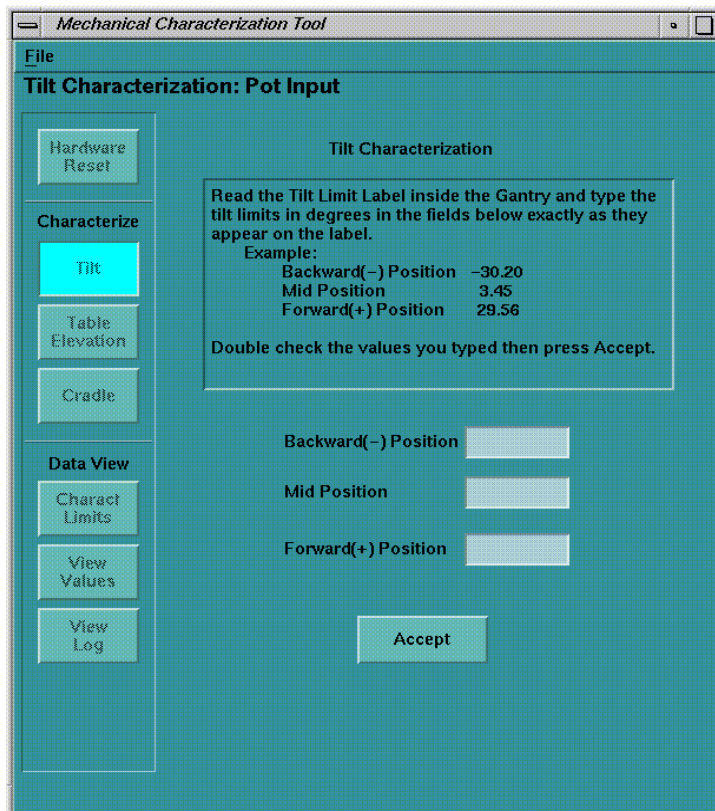
3.) Select INSTALL.

4.) Select TABLE GANTRY INTEGRATION.

5.) Select CHARACTERIZATION.

6.) Select TILT.

7.) Follow the Tilt Characterization instructions on the screen.



The screenshot shows the 'Mechanical Characterization Tool' window. The title bar reads 'Mechanical Characterization Tool'. The main window has a menu bar with 'File'. Below the menu bar, the title 'Tilt Characterization: Pot Input' is displayed. On the left side, there is a vertical toolbar with buttons: 'Hardware Reset', 'Characterize' (which is highlighted), 'Tilt' (which is highlighted in red), 'Table Elevation', 'Cradle', 'Data View', 'Charact Limits', 'View Values', and 'View Log'. The main area of the window is titled 'Tilt Characterization'. It contains a text box with instructions: 'Read the Tilt Limit Label inside the Gantry and type the tilt limits in degrees in the fields below exactly as they appear on the label. Example: Backward(-) Position -30.20, Mid Position 3.45, Forward(+) Position 29.56. Double check the values you typed then press Accept.' Below this text box are three input fields: 'Backward(-) Position', 'Mid Position', and 'Forward(+) Position'. At the bottom right of the main area is an 'Accept' button.

Figure A-1 Gantry Tilt Characterization Screen

Section 2.0

Elevation Characterization

2.1 Prerequisite

The table to gantry MUST be level before performing this procedure. The following tools are required:

- Metric tape measure
- Flat blade screwdriver
- Metric hex key set
- 9" level

2.2 Procedure

- 1.) Remove upper right and lower right table covers.
- 2.) Locate the small L-shaped measurement block, stored on top of the bigger measurement block.
 - a.) Remove the small L-shaped measurement block from its storage position.
 - b.) Install it in its characterization position, at the right rear of the table.
 - c.) Move the measurement plate, on the right side of table, to the **out** position.
 - d.) Tighten the mounting screws for both plates.

- 3.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 4.) Click on the UTILITIES icon.



- 5.) Select INSTALL.

- 6.) Select TABLE GANTRY INTEGRATION.

- 7.) Select CHARACTERIZATION.

- 8.) Select TABLE ELEVATION.

- 9.) Follow the Elevation Characterization instructions displayed on the screen. *Important - Take care to measure the exact elevation distances. Incorrect measurement results in Elevation Characterization failure.* Table Low = 374.1 mm. Table High = 849.1 mm.

Refer to [Figure A-2](#). Measure the elevation distance as vertically as possible. Measure between the bottom surface of the each measurement block and the center of the rear leg upper pivot pin.

- 10.) Return the measurement plates to the storage position.

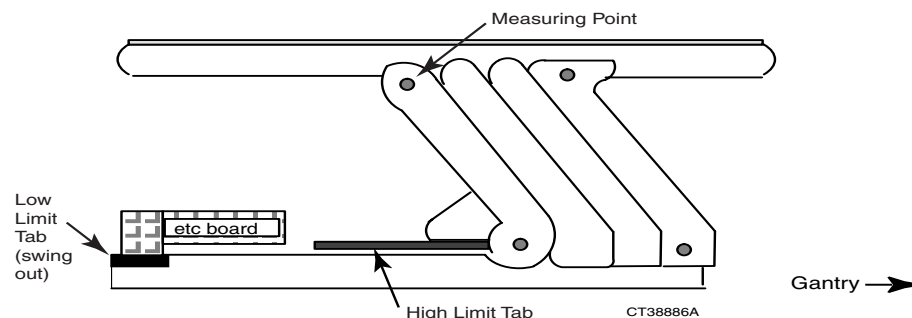
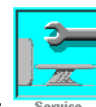


Figure A-2 Measure Elevation Distance

Section 3.0

Cradle Characterization



- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select TABLE GANTRY INTEGRATION.
- 5.) Select CHARACTERIZATION.
- 6.) Select CRADLE.
- 7.) Follow the cradle characterization instructions displayed on the screen.

Note:
A table/gantry
reset may be
required.

If the display flashes "888", then you must reset the table before proceeding. To reset the table, lower it to its lowest position and then raise it to its maximum position.

- a.) In the alignment light procedure (3rd and 4th screens), use the BLACK DOT on the cradle right-side bumper.

Elevate the table to at least 40mm from isocenter, so the black dot reaches the external scan plane alignment light *before* the cradle reaches its home position.

- b.) In the end-of-cradle position procedure (5th screen), use the same BLACK DOT on the cradle right-side bumper.

Note: **Do not use buttons to drive table cradle.**

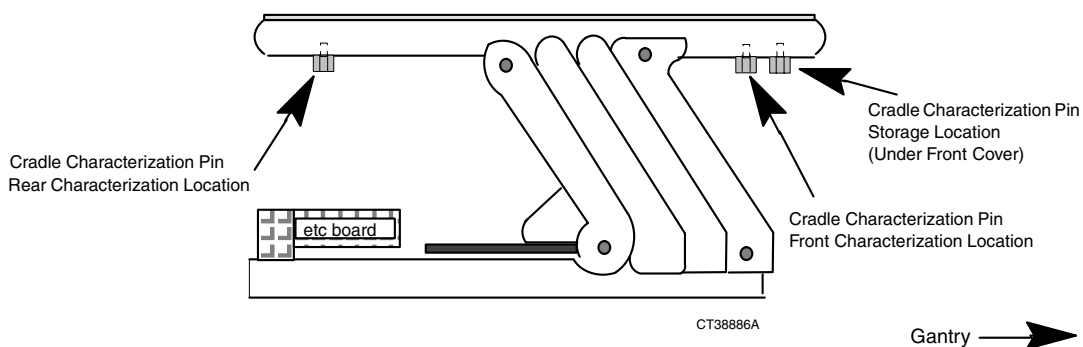


Figure A-3 Cradle Characterization Pin Locations

Section 4.0

Tilt Speed Adjustment Procedure

- 1.) Remove rear gantry base covers.
- 2.) Using a stop watch or suitable time piece adjust for one (1) degree per second motion.



NOTICE Adjustments are very sensitive. Improper speed adjustments will result in overspeed error generation.

Note: Tilt speed will vary based upon hydraulic fluid temperature. Adjustments should be made at normal scan room temperature settings.

- 3.) From zero (0) degrees tilt, press and hold the tilt forward button,  until gantry display reads S25. Observe tilt speed using time piece.
- 4.) Press and hold backward tilt button,  until gantry display reads I25. Observe tilt speed using time piece.
- 5.) Adjust forward speed control valve for one (1) degree per second motion. See [Figure 6-1](#).
- 6.) Adjust backward speed control valve for one (1) degree per second motion. See [Figure 6-1](#).
- 7.) Repeat steps 3 through 6 until tilt speed for both directions is correct.
- 8.) Now tilt the gantry forward to S30. Observe the S25 to S30 speed. Do this several times. If a noticeable difference is observed, then check the hydraulic fluid levels.

Note: This is a self bleeding system. Trapped air can cause slowed or limited tilt range. Exercising the full range of motion several times should purge any trapped air from the hydraulic system.

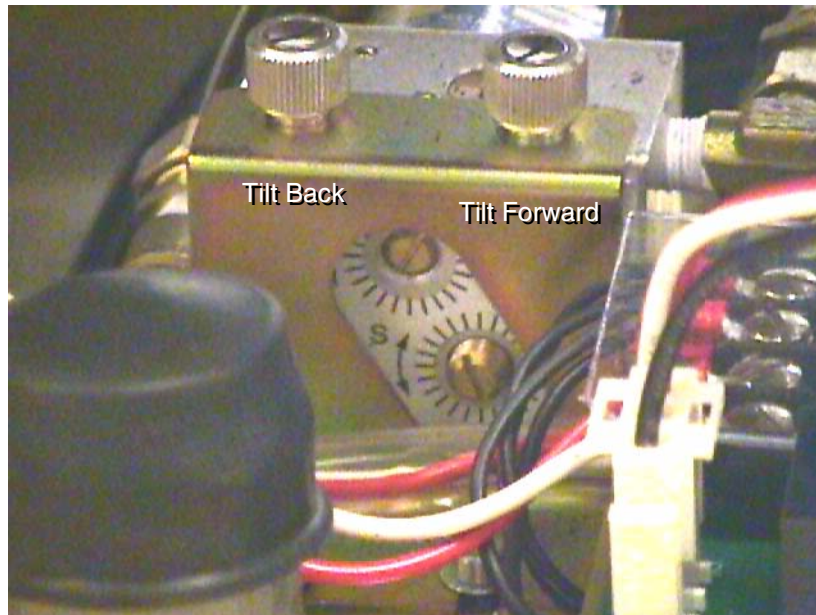


Figure 6-1 Tilt Speed Adjustment Screws

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Waukesha, Wisconsin 53188
USA

www.gehealthcare.com

